

Grishma Gajurel
Webster University
Fall 2025

Executive Summary

North Point Software, a specialized firm in the gaming and educational software industry, is employing mailing analytics to enhance its marketing strategies and expand its customer base. It is seeking to improve the effectiveness of its mailing campaigns by better understanding customer purchase behavior. The dataset under review contains mailing sources alongside information about how many individuals were contacted, how many of them purchased, and the corresponding purchase rates. This information provides insight into which sources generate stronger engagement and conversions, highlighting that not all customer segments or mailing lists perform equally. By examining these differences, the dataset supports the company's broader business goal of optimizing marketing spend, minimizing wasted outreach, and targeting customers more likely to convert.

The primary business challenge lies in the inefficiencies of broad, untargeted mailing campaigns, which frequently result in low purchase rates, increased costs, and diminished profitability. North Point's objectives are twofold: first, to augment profitability by accurately classifying which customers are likely to make purchases, and second, to optimize marketing efficiency by segmenting customers based on their purchase behaviors and engagement characteristics. By comprehending the correlation between attributes such as spending, purchase frequency, and the acquisition sources of customers, the company can make informed decisions regarding which segments to prioritize in future mailings. This ensures that resources are allocated toward customers with the highest potential for conversion and long-term value.

From an analytical standpoint, segmentation serve as pathways to achieving these objectives. Classification models facilitate the prediction of purchase likelihood for each customer, supporting decision-making in selecting names from the consortium's pooled list. Unsupervised methods further enhance comprehension by grouping customers into meaningful segments that unveil concealed patterns in purchasing tendencies and source performance. These analytical approaches collectively constitute a robust framework for addressing both short-term profitability and long-term customer loyalty.

An important consideration when applying models to this dataset is the adjustment required for probability estimates. Because the dataset used in modeling does not reflect the true population purchase rate, predicted probabilities of purchase must be scaled down to more realistic levels. Specifically, after predicting which cases are likely to be purchasers, the purchase rate must be adjusted back down by multiplying each case's probability of purchase by $5.3/50$, or 0.106 . This ensures that forecasts are not overly optimistic and remain consistent with the actual business environment, where only a small fraction of the total audience converts.

Eventually, the success of North Point Software's mailing strategy hinges on its capacity to harness data-driven insights to guide customer selection and campaign design. The dataset is ideally suited for this purpose, encompassing both behavioral and demographic indicators that can be directly correlated with business outcomes. This alignment of business objectives with analytical methodologies positions North Point to maximize returns from its marketing investments, strengthen customer relationships, and maintain competitiveness in a competitive software market.

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1. Introduction

North-Point, a software firm specializing in games and educational software, has expanded its offerings to include third-party titles, diversifying its catalog over time. In addition to products, North-Point's customer list is an asset that supports direct marketing initiatives. To further grow its customer base, North-Point joined a group of listing firms specializing in computer hardware and software. This collaboration allowed members to exchange customer lists and conduct predictive modeling for targeted marketing.

As part of this effort, North-Point contributed 200,000 customer names to a shared pool of over 5 million names, enabling it to withdraw an equivalent number for direct mail campaigns. A test mailing of 20,000 names yielded 1,065 purchasers (response rate = 5.3%), providing a dataset for predictive modeling. To optimize machine learning models, a stratified sample of 1,000 purchasers and 1,000 non-purchasers was constructed (apparent response rate = 50%). Predictions must subsequently be adjusted to reflect the true baseline response rate (5.3%).

The central challenge is to identify individuals most likely to purchase from the mailing list, thereby maximizing campaign profitability and minimizing wasted marketing expenditure.

2. Business & Analytical Goals

2.1 Business Problem

- **Inefficient Customer Targeting:** Randomly selecting from the mailing pool risks sending promotions to uninterested individuals, diminishing ROI and leading to wasted campaign efforts.
- **Budget Constraints:** Direct mail campaigns involve significant costs in printing, postage, and handling. Without effective targeting, these costs reduce overall profitability and place financial strain on marketing budgets.
- **Customer Acquisition Uncertainty:** Without accurate predictive modeling, North Point risks attracting customers with low lifetime value who do not significantly contribute to long-term revenue growth.
- **Over-Saturation Risk:** Repeatedly mailing the same broad audience without personalization can lead to customer fatigue, decreased engagement, and potential damage to brand perception.
- **Lack of Segment-Specific Insights:** Treating the entire customer base as uniform prevents North Point from identifying subgroups with unique behaviors, spending patterns, and responsiveness. This results in missed opportunities for tailored offers and relationship building.
- **Ineffective Measurement of Campaign Success:** Without advanced predictive models, North Point relies heavily on raw response counts. This limits the ability to evaluate which factors truly drive conversions and makes it harder to refine future campaigns.

2.2 Business Goals

Goal 1: Improve targeting Precision

Random targeting results in wasted mailing costs and diminished return of investment. By employing certain models, North Point can forecast the probability of purchase for each customer and prioritize high-probability responders. This strategy ensures that marketing resources are concentrated on individuals most likely to convert, thereby enhancing efficiency, minimizing expenses, and directly addressing the issue of inefficient customer targeting.

- Implement a strategy that achieves a minimum 20% improvement in response rate, surpassing the effectiveness of random targeting.
- Reduce mailing costs per conversion by 15–20% through targeted distribution.
- Enhance the return on investment of direct mail campaigns by a minimum of 10% through data-driven targeting strategies.

Goal 2: Discover Customer Segments

Customers in the mailing pool are not uniform — they differ in location, frequency, and responsiveness. Some techniques can uncover distinct customer groups based on transaction history, web activity, or spending behavior. These segments allow North Point to tailor campaign strategies, create differentiated offers, and strengthen long-term engagement. This approach not only enhances campaign relevance but also builds a foundation for sustained customer relationship management.

- Identify at least three to five actionable customer segments with statistically distinct characteristics.
- Implement segment-specific marketing strategies that enhance engagement and mitigate over-saturation.
- Enhance long-term retention or repeat purchase rates by 10% through personalized communication strategies.

2.3 Analytical Goals

Predict Purchase Probability

- Develop predictive models (Logistic Regression, Neural Network) to estimate the probability of purchase for each customer.
- Approach:
 - 1. Model Development: Train predictive models to assess the likelihood of purchase for individual customers.
 - 2. Probability Calibration: Adjust the estimated probabilities to align with the actual response rate of 5.3%. This adjustment will be applied using the 0.106 adjustment factor.
 - 3. Ranked Customer List: Generate a ranked customer list based on the predicted purchase probabilities. This list will serve as a guide for making informed mailing decisions.

Identify Key Predictors of Purchase Behavior

- Utilize model coefficients and feature importance (derived from Logistic Regression, Decision Trees, or Neural Networks) to identify the most significant customer attributes.
- Outcome: A comprehensive data-driven conception of the factors influencing customer responses, thereby informing future offer development and audience selection strategies.

Uncover Natural Customer Segments

- Apply Clustering Algorithms: Utilize K-Means and Hierarchical clustering algorithms to segment customers based on frequency, spending patterns, and engagement levels.
- Outcome: Define well-defined customer profiles that facilitate personalized marketing strategies and mitigate campaign fatigue.

Evaluate and Validate Model Performance

- Assessing Model Reliability: Utilize metrics such as accuracy, precision, recall, etc., to evaluate the reliability of the model.
- Measuring Business Impact: Employ ROC/AUC charts to quantify the business impact of the model and ensure that the predicted probabilities align with actual outcomes.
- Outcome: A validated, business-ready predictive framework for campaign targeting and measurement.

2.4 Analytical Approach

To address business challenges and attain North Point's objectives, a multi-layered approach employing predictive and descriptive modeling techniques will be adopted. Each algorithm is selected strategically based on its capacity to address specific challenges effectively and contribute to campaign optimization.

Phase 1: Predictive Modeling for Targeting

Logistic Regression

Logistic regression, selected for its interpretability, quantifies the influence of customer attributes on the likelihood of a purchase.

- Addresses: Inefficient targeting and budget constraints can be mitigated by identifying high-probability responders, facilitating focused mailings and cost reductions.
- Outcome: A prioritized customer list that optimizes campaign allocation.

Neural Networks

Deployed to capture complicated nonlinear relationships that conventional models may overlook.

- Addresses: Customer acquisition uncertainty by improving predictive accuracy and identifying high-value prospects that logistic regression may overlook.
- Outcome: Enhanced targeting precision and scalable prediction performance.

Phase 2: Customer Segmentation and Insights

Clustering Techniques (K-Means, Hierarchical)

Employed to identify inherent groupings predicated on behavioral and transactional attributes.

- Addresses: Limited segment-specific insights and over-saturation risk due to the revelation of distinct behavioral segments for tailored engagement.
- Outcome: Actionable customer profiles that underpin personalized marketing strategies.

Phase 3: Campaign Measurement and Model Validation

Evaluation Metrics

Model performance evaluation will be conducted utilizing metrics such as accuracy, precision, recall, and others.

- Addresses: Ineffective campaign measurement by providing objective performance evaluation even in the presence of class imbalance.
- Outcome: Robust models that faithfully replicate purchasing patterns.

ROC Curve and AUC (Area Under the Curve)

Utilized to assess the model's discriminatory ability in differentiating between prospective buyers and non-buyers.

- Addresses: Quantifying model reliability by assessing the extent to which predictions distinguish responders from non-responders.
- Outcome: The high AUC values demonstrate robust predictive performance and enhanced targeting precision.

Probability Calibration

Predicted probabilities will be aligned with the actual purchase rate (by multiplying by 0.106).

- Addresses: Overestimation risk in acquisition planning by ensuring that predicted probabilities accurately reflect actual response rates.
- Outcome: Realistic, business-ready probability estimates for campaign decision-making.

3. Data Preprocessing

3.1 Data Understanding

The dataset provided by North-Point comprises a stratified sample of 2,000 customer records, drawn from the broader mailing experiment that encompassed 20,000 participants. This balanced dataset comprises 1,000 purchasers and 1,000 non-purchasers, facilitating effective model training while acknowledging that the actual purchase rate in the test mailing was 5.3%. Following model development, the predicted probabilities of purchase must be rescaled to align with this actual rate.

3.2 Attribute Definitions

1. sequence_number

- Description: A unique identifier for each customer record.
- Type: Numerical (Index)
- Business Relevance: Internal tracking and record management.
- Analytical Relevance: Not useful for predictive modeling; excluded from analysis.

2. US

- Description: This indicator determines whether the customer is in the United States (1 = Yes, 0 = No).
- Type: Binary.
- Business Relevance: U.S.-based customers are more receptive to faster delivery and localized messaging, which aligns with North Point's objective of enhancing domestic marketing efficacy.
- Analytical Relevance: Allows for the segmentation of domestic and international response patterns, facilitating geographically tailored campaigns.

3. source_ (15 binary variables)

- Description: This field represents the marketing source or channel from which the customer record originated.
- Type: Binary.
- Business Relevance: Some channels may yield higher-quality leads than others.
- Analytical Relevance: This field has high potential for predictive modeling; it can assist in identifying profitable acquisition sources.

4. Freq

- Description: The number of transactions conducted by customers through catalogs within the past year.
- Type: Numerical
- Business Relevance: Ensures customer loyalty and engagement, thereby facilitating the prioritization of repeat customers who significantly contribute to revenue.
- Analytical Relevance: A crucial factor in purchase and profitability models; facilitates targeted retention and cross-selling strategies.

5. last_update_days_ago

- Description: The number of days since the customer record was last updated.
- Type: Numerical
- Business Relevance: The recency of updates often indicates more recent and accurate records.
- Analytical Relevance: A measure of recency.

6. 1st_update_days_ago

- Description: The number of days since the initial update to the customer record.
- Type: Numerical
- Business Relevance: This variable indicates the customer's tenure in the database.
- Analytical Relevance: It may reflect long-term customer engagement and is useful for distinguishing loyal, long-term customers from newer entries.

7. Web_order

- Description: This indicator determines whether the customer has previously placed an online order (1 = Yes, 0 = No).
- Type: Binary
- Business Relevance: It suggests digital readiness and a preference for convenience.
- Analytical Relevance: This categorical predictor indicates a higher purchase likelihood among customers with prior web orders.

8. Gender=male

- Description: Gender indicator (1 = Male, 0 = Female/Other).
- Type: Binary.
- Business Relevance: Allows for the customization of campaign content and product recommendations based on gender-associated preferences.
- Analytical Relevance: Acts as a demographic segmentation variable that helps personalization.

9. Address_is_res

- Description: Indicates whether the address is residential (1 = Yes, 0 = No).
- Type: Binary.
- Business Relevance: Residential addresses may yield higher direct mail responsiveness.
- Analytical Relevance: Utilizes geographic and behavioral segmentation to enhance targeting precision and minimize mailings that are not delivered.

10. Purchase

- Description: Response variable indicating whether the customer made a purchase in the test mailing (1 = Yes, 0 = No).
- Type: Binary
- Business Relevance: Directly correlated with campaign ROI.
- Analytical Relevance: Primary classification target; models predict the probability of purchase.

11. Spending

- Description: The total monetary value spent by customers during the test mailing campaign.
- Type: Numerical
- Business Relevance: This variable provides insights into the customer's financial value beyond their purchase response.
- Analytical Relevance: It serves as a regression target for revenue prediction, enabling profitability-driven targeting strategies.

3.3 Missing and Zero Data Analysis

sequence_number	US	source_a	source_c	source_b	source_d	source_e
Min. : 1.0	Min. :0.0000	Min. :0.0000	Min. :0.000	Min. :0.00	Min. :0.0000	Min. :0.000
1st Qu.: 500.8	1st Qu.:1.0000	1st Qu.:0.0000	1st Qu.:0.000	1st Qu.:0.00	1st Qu.:0.0000	1st Qu.:0.000
Median :1000.5	Median :1.0000	Median :0.0000	Median :0.000	Median :0.00	Median :0.0000	Median :0.000
Mean :1000.5	Mean :0.8245	Mean :0.1265	Mean :0.056	Mean :0.06	Mean :0.0415	Mean :0.151
3rd Qu.:1500.2	3rd Qu.:1.0000	3rd Qu.:0.0000	3rd Qu.:0.000	3rd Qu.:0.00	3rd Qu.:0.0000	3rd Qu.:0.000
Max. :2000.0	Max. :1.0000	Max. :1.0000	Max. :1.000	Max. :1.00	Max. :1.0000	Max. :1.000
source_m	source_o	source_h	source_r	source_s	source_t	source_u
Min. :0.0000	Min. :0.0000	Min. :0.0000	Min. :0.0000	Min. :0.000	Min. :0.0000	Min. :0.000
1st Qu.:0.0000	1st Qu.:0.0000	1st Qu.:0.0000	1st Qu.:0.0000	1st Qu.:0.000	1st Qu.:0.0000	1st Qu.:0.000
Median :0.0000	Median :0.0000	Median :0.0000	Median :0.0000	Median :0.000	Median :0.0000	Median :0.000
Mean :0.0165	Mean :0.0335	Mean :0.0525	Mean :0.0685	Mean :0.047	Mean :0.0215	Mean :0.119
3rd Qu.:0.0000	3rd Qu.:0.0000	3rd Qu.:0.0000	3rd Qu.:0.0000	3rd Qu.:0.000	3rd Qu.:0.0000	3rd Qu.:0.000
Max. :1.0000	Max. :1.0000	Max. :1.0000	Max. :1.0000	Max. :1.000	Max. :1.0000	Max. :1.000
source_p	source_x	source_w	Freq	last_update_days_ago	1st_update_days_ago	Web.order
Min. :0.000	Min. :0.000	Min. :0.0000	Min. : 0.000	Min. : 1	Min. : 1	Min. :0.000
1st Qu.:0.000	1st Qu.:0.000	1st Qu.:0.0000	1st Qu.: 1.000	1st Qu.:1133	1st Qu.:1671	1st Qu.:0.000
Median :0.000	Median :0.000	Median :0.0000	Median : 1.000	Median :2280	Median :2721	Median :0.000
Mean :0.006	Mean :0.018	Mean :0.1375	Mean : 1.417	Mean :2155	Mean :2436	Mean :0.426
3rd Qu.:0.000	3rd Qu.:0.000	3rd Qu.:0.0000	3rd Qu.: 2.000	3rd Qu.:3139	3rd Qu.:3353	3rd Qu.:1.000
Max. :1.000	Max. :1.000	Max. :1.0000	Max. :15.000	Max. :4188	Max. :4188	Max. :1.000
Gender.male	Address_is_res	Purchase	Spending			
Min. :0.0000	Min. :0.000	Min. :0.0	Min. : 0.0			
1st Qu.:0.0000	1st Qu.:0.000	1st Qu.:0.0	1st Qu.: 0.0			
Median :1.0000	Median :0.000	Median :0.5	Median : 2.0			
Mean :0.5245	Mean :0.221	Mean :0.5	Mean : 102.6			
3rd Qu.:1.0000	3rd Qu.:0.000	3rd Qu.:1.0	3rd Qu.: 153.0			
Max. :1.0000	Max. :1.000	Max. :1.0	Max. :1500.0			

Figure 1: Summary of the dataset

The dataset does not contain any explicit missing values (NA) reported in the summary output. All attributes have predefined minimums and maximums. The presence of zeros across multiple attributes suggests that they are binary (0/1). Continuous attributes such as spending, last_update_days_ago, and 1st_update_days_ago exhibit a wide range of values without any indication of missingness but may exhibit potential skewness.

3.4 Exploratory Data Analysis (Basic)

3.4.1 Continuous Variable Analysis

The sequence_number is not included in the description because it is an index.

	vars	n	mean	sd	median	trimmed	mad	min	max	range	skew	kurtosis	se
sequence_number	1	2000	1000.50	577.49	1000.5	1000.50	741.30	1	2000	1999	0.00	-1.20	12.91
Freq	2	2000	1.42	1.41	1.0	1.22	1.48	0	15	15	2.98	15.98	0.03
last_update_days_ago	3	2000	2155.10	1141.30	2280.0	2185.11	1490.01	1	4188	4187	-0.19	-1.19	25.52
X1st_update_days_ago	4	2000	2435.60	1077.87	2721.0	2507.64	1165.32	1	4188	4187	-0.49	-0.81	24.10
Spending	5	2000	102.62	186.78	2.0	63.46	2.97	0	1500	1500	3.92	20.50	4.18

Figure 2: Detailed Descriptive summary

1. Freq

- Mean: 1.42
- Median: 1
- Standard Deviation: 1.41
- Distribution: The distribution exhibits a strong right skew, characterized by a high proportion of values to the right of the median.
- Business Relevance: The prevalence of one-time buyers underscores cruciality for retention-centric strategies. Frequent customers constitute a lucrative loyalty segment that should be prioritized for personalized offers and exclusive access promotions.

2. last_update_days_ago

- Mean: 2155 days
- Median: 2280 days
- Standard Deviation: 1141
- Distribution: The distribution exhibits a slight left skew (Skew = -0.19).
- Business Relevance: The prevalence of outdated records suggests potential data quality issues that could compromise targeting accuracy. Conversely, records that have been recently updated are likely to represent more engaged and accessible customers.

3. 1st_update_days_ago

- Mean = 2436 days
- Median = 2721 days
- Standard Deviation = 1078
- Distribution: The distribution exhibits a slight left skew.
- Business Relevance: Customer retention management involves capturing and retaining loyal customers. Long-term customers can be utilized for loyalty programs, while newer customers may benefit from introductory campaigns.

4. Spending

- Mean = 102.6
- Median = 2
- Standard Deviation = 186.8
- Distribution: Extremely right skewed, with a long tail.
- Business Relevance: Spending is concentrated among a select segment of customers. This insight can guide tiered marketing strategies: low-spending customers necessitate conversion offers, while high-spending customers require personalized retention and loyalty initiatives.

3.4.2 Purchase Proportions Analysis

Purchase Proportion by Binary Variables

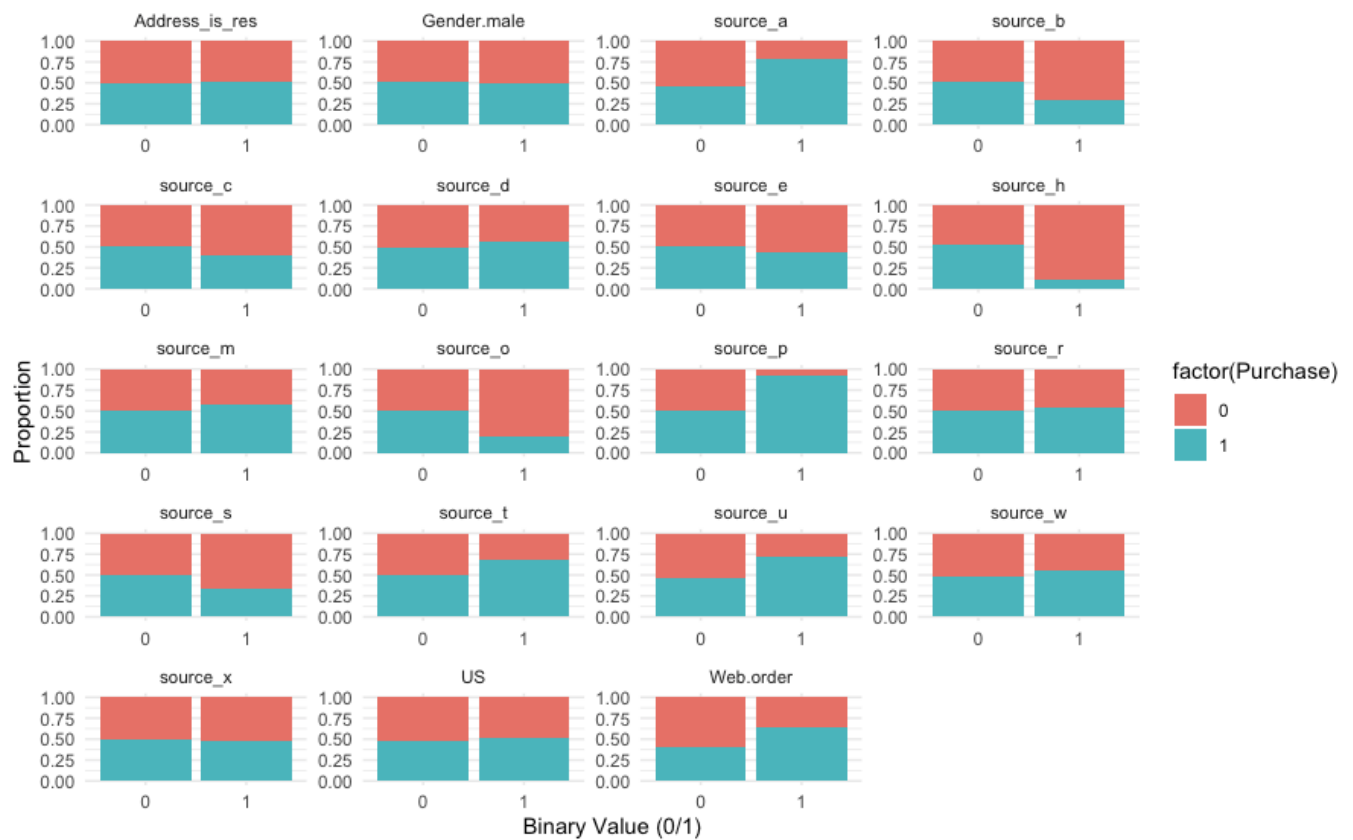


Figure 3: Purchase Proportion of Binary Variables

Each subplot corresponds to a binary variable (0/1). The x-axis represents the values “No” for that attribute and “Yes” respectively. The y-axis displays the proportion of customers. The blue (bottom) segment represents the proportion of customers who made a purchase (Purchase = 1), while the red (top) segment represents the proportion of customers who did not purchase (Purchase = 0). This visualization illustrates how purchase behavior varies between the two groups (0 vs 1) of each binary variable.

Detailed Variable Insights

1. Address_is_res

- Customers with a residential address (1) and those without (0) exhibit nearly identical purchase proportions.
- Business relevance: Address type may not significantly influence purchase decisions; nevertheless, verifying identity and residency could still aid in fraud detection.
- Analytical relevance: There is no discernible separation in purchase rates; therefore, it lacks predictive value.

2. Gender.male

- Male and non-male shows exhibit comparable purchase proportions.
- Business relevance: Gender segmentation may not yield significant value in this targeting context.
- Analytical relevance: Gender is a weak predictor when used alone. Its value is enhanced by interaction effects with other features.

3. Source variables (source_*)

- Source a, b, c, d, m, r, x: A moderate variation in purchase proportions, but not an extreme one.
- Source e, h, p, s, t, u, w: show sharper contrasts. Example:
 - Source p (1): Customers from this channel exhibit a significantly higher purchasing rate (blue dominates).
 - Source h (1): Customers from this channel demonstrate a substantially lower purchasing rate (red dominates).
- Business relevance: High-Conversion Channels: Channels with high conversion rates, such as source_p, should be prioritized for marketing spend allocation. Channels with low conversion rates, like source_h, indicate potential misaligned targeting.
- Analytical relevance: Sources such as p, u, and t will be strong predictors in modeling purchase likelihood. Channel analysis enables profit optimization.

4. US

- US and Non-US Customers: Proportions are balanced, with slight variations.
- Business relevance: Domestic vs. international status does not significantly impact purchase probability.
- Analytical relevance: This variable alone lacks strong predictive power. However, when combined with the marketing source, it may provide valuable insights (e.g., US + source_p = high purchase as the product can be shipped quickly and easily).

5. Web.order

- Customers ordering through the web exhibit a slightly higher purchase proportion compared to those who do not.
- Business relevance: This observation suggests that digital channels are marginally more effective than other methods. Therefore, it strengthens the case for investing in online campaigns.
- Analytical relevance: The predictive power is moderate, but it aligns with customer convenience trends.

3.4.3 Correlation Plot for Multicollinearity

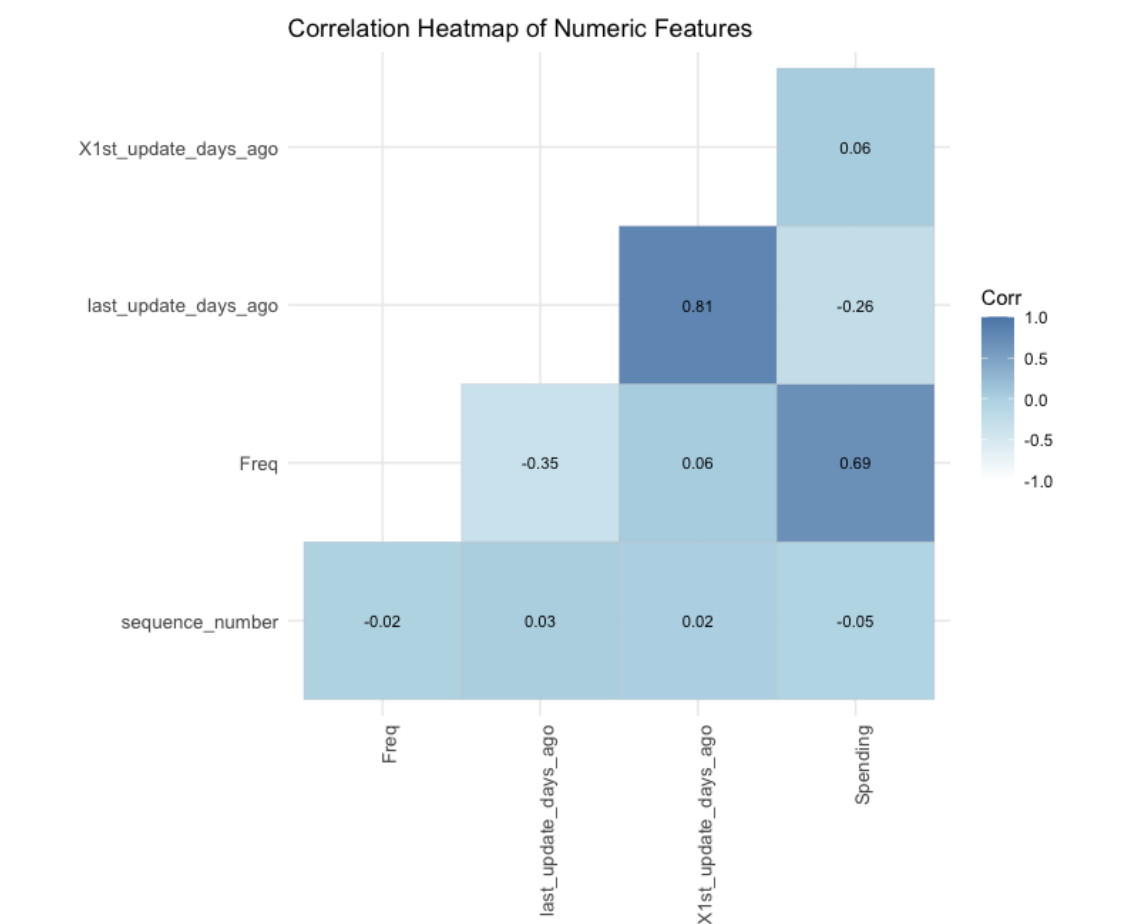


Figure 4: Correlation plot of numeric variables

The correlation heatmap of numerical features reveals statistically significant and strategically meaningful relationships for profitability prediction. A strong correlation (0.81) between `last_update_days_ago` and `1st_update_days_ago` suggests redundancy and potential multicollinearity, indicating that customers with older initial records also tend to have outdated recent updates. From a campaign strategy perspective, this highlights the importance of data freshness—customers with stale or infrequently updated records are typically disengaged, making them low-priority targets for future mailings.

The positive correlation (0.69) between frequency and spending reinforces that loyal, repeat customers are key revenue drivers. These features jointly serve as powerful indicators for identifying high-value segments, guiding both regression-based profitability forecasting and targeted retention campaigns. Conversely, the negative relationship (-0.35) between frequency and `last_update_days_ago` implies that recently active customers are also frequent purchasers, underscoring the value of real-time behavioral tracking in campaign timing.

While variables like `sequence_number` lack clear predictive or strategic value, the collective insights from spending and transaction frequency directly inform North Point's business objective of maximizing profitability through smarter segmentation and resource allocation. By aligning data-driven predictors with customer engagement strategies, the company can refine its targeting, reduce marketing inefficiencies, and strengthen its ROI.

4. Predictor Analysis and Relevancy

4.1 Analysis and Relevancy

Logistic Regression (Purchase Prediction)

- Best Predictors:
 - Freq -> Captures customer activity, which is highly correlated with purchase probability.
 - last_update_days_ago / 1st_update_days_ago -> Time-based predictors are useful for odds interpretation and can be utilized as random effects. (recency strongly affects response).
 - Gender, Address_is_res, US -> Categorical or binary predictors that logistic regression effectively handles, facilitating straightforward interpretation of coefficients.
- Logistic regression excels in interpreting binary or numerical predictors, making it an effective tool for understanding the influence of each factor on the probability of purchase.
- Variables with minimal statistical contribution provide limited incremental business insights, as they fail to distinguish active customers from inactive ones. Handling them simplifies the model, ensuring that predictive insights remain actionable rather than being obscured by noise.

Neural Network (Complex, Nonlinear Relationships)

- Best Predictors:
 - Spending -> A continuous variable with nonlinear thresholds.
 - Freq -> Interactions with Spending may generate nonlinear purchase likelihood patterns.
 - Source_ (15 source lists) * -> Neural networks excel at capturing interactions across multiple sources, which logistic regression may overlook due to the presence of many binary predictors.
- Neural networks effectively manage complicated, nonlinear interactions among predictors.
- It offers enhanced capabilities to discern concealed interactions between behavioral and demographic variables that logistic regression may overlook. This facilitates refined targeting and personalized campaign design, particularly in scenarios where nonlinear response patterns are present.

K-Means Clustering (Customer Segmentation)

- Best Predictors:
 - Spending -> Identifies and categorizes customers based on their perceived value.
 - Freq -> Distinguishes frequent customers from occasional customers.
 - last_update_years / 1st_update_years -> Segment customers based on their recency.
- Clustering requires the utilization of continuous variables to categorize customers based on behavioral similarity. Therefore, Spending, Frequency, and Updates emerge as the most potent inputs for the description of meaningful customer segments.
- Redundant or categorical variables contribute minimal segmentation value in numerical clustering frameworks. Their exclusion enhances interpretability and concentrates the segmentation on customer behavior, directly aligning with Goal 2: Identify Customer Segments.

Hierarchical Clustering

- Best Predictors:
 - Source_ (15 source lists) * -> Displays a natural clustering of customers based on their acquisition channel.
 - US, Gender, Address_is_res -> Categorical or binary features effectively identify natural subgroups.
- Hierarchical clustering is a suitable method for identifying nested customer groups.
- It is particularly valuable for visualizing nested relationships, such as identifying the differences in channel responsiveness between domestic and international segments. This hierarchy facilitates tiered marketing strategies, ranging from broad demographic targeting to niche, source-specific campaign personalization.

4.2 Handling Multicollinearity

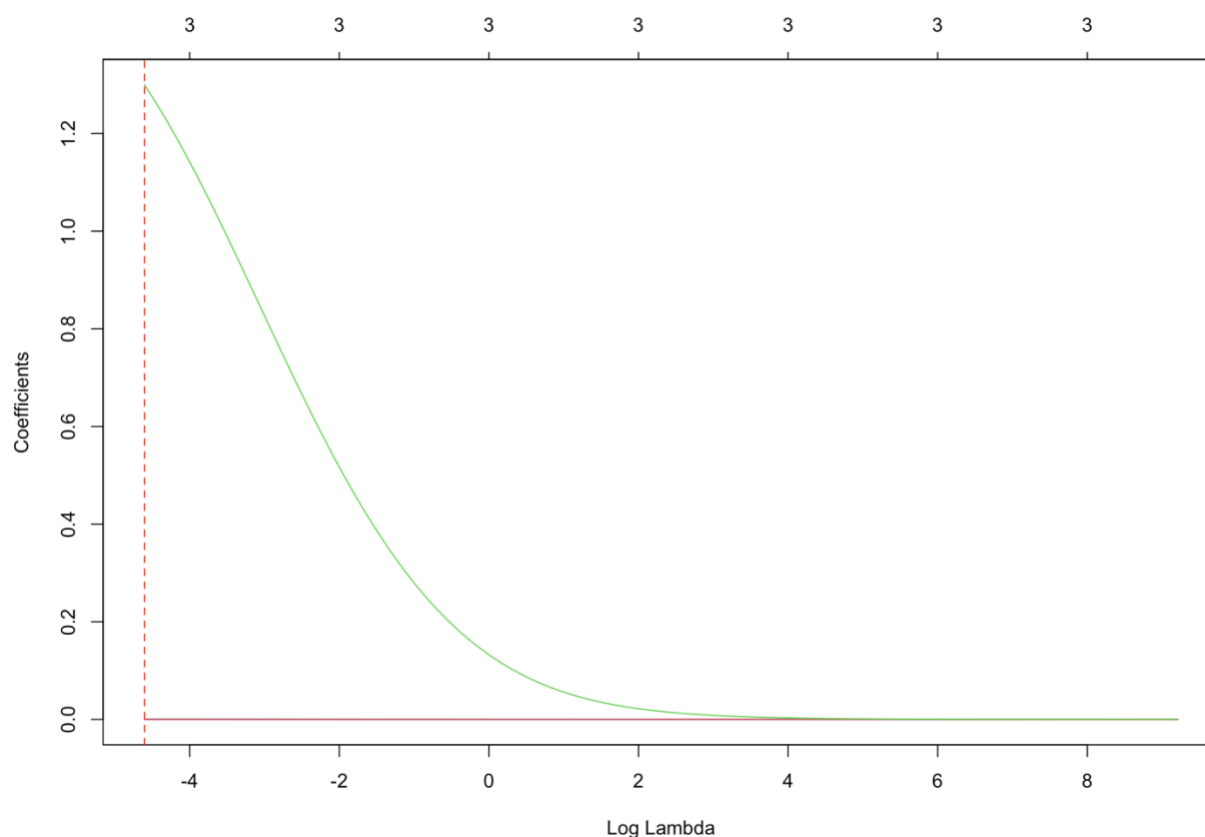


Figure 5: Ridge coefficient plot

The ridge regression coefficient shrinkage path offers valuable insights into the statistical behavior of predictors and their strategic relevance to North Point's mailing campaign. The graph illustrates how the coefficients of predictors such as `X1st_update_days_ago`, `last_update_days_ago`, and `Freq` evolve as the regularization parameter increases. At lower lambda values, the coefficients closely resemble their original regression estimates, reflecting the full influence of each variable. As lambda increases, the coefficients gradually shrink toward zero—never eliminated but balanced in proportion—demonstrating how ridge regression distributes influence more equitably among correlated predictors.

From an analytical perspective, this regularization directly addresses the strong correlation observed between `X1st_update_days_ago` and `last_update_days_ago`, which could otherwise distort coefficient estimates and lead to erroneous conclusions about customer behavior. Ridge regression mitigates this risk by shrinking correlated predictors in tandem, preventing redundancy from overstating the impact of outdated customer data. In contrast, independent and behaviorally meaningful variables such as `Freq` retain greater stability, reinforcing their dominant role in predicting purchase likelihood.

From a business standpoint, ridge regression not only resolves multicollinearity but also enhances decision reliability. By stabilizing predictor relationships, it ensures that probability estimates of purchase behavior remain consistent and credible even when variables overlap. This translates into more precise customer targeting, enabling North Point to allocate mailing resources toward customers most likely to respond while minimizing expenditures on low-probability segments. In essence, ridge regression supports more informed investment decisions by preventing overreliance on any single predictor and ensuring that each factor contributes proportionally to campaign outcomes.

Furthermore, ridge regularization not only enhances interpretability but also holds strategic significance. It empowers marketing teams to comprehend the relative importance of behavioral and temporal predictors, thereby clarifying, for instance, that frequently and recently updated customers consistently constitute high-value targets. This clarity facilitates data-driven prioritization of customer segments, thereby enhancing targeting precision and overall campaign return on investment.

5. Data Engineering and Transformation

5.1 Column Conversion and Segmentation

last_update_years	X1st_update_years	last_update_segment	first_update_segment
10.032877	10.032877	10-11 years	10-11 years
7.945205	7.945205	7-8 years	7-8 years
10.638356	10.723288	10-11 years	10-11 years
2.271233	2.271233	2-3 years	2-3 years
2.380822	2.380822	2-3 years	2-3 years
5.465753	5.484932	5-6 years	5-6 years

Figure 6: Transformation of predictors

A transformative shift occurred when time-related attributes, initially measured in days, were converted into years. This transformation enhances interpretability for both analysts and marketing stakeholders, as yearly intervals are more relatable when assessing customer engagement cycles. For instance, knowing that a customer last updated their information three years ago conveys a clearer business meaning compared to 1,095 days ago. This temporal framing enables North Point to view customer activity in practical marketing terms and design targeted reactivation or retention strategies accordingly.

To further align data with actionable insights, these time-based variables were segmented into categorical bins (e.g., 0–1 year, 1–2 years, up to 11 years). This segmentation transforms continuous time data into discrete behavioral categories that mirror real-world marketing segments. For example, customers who updated within the past year may represent loyal or recently engaged individuals suitable for loyalty offers, while those inactive for over five years may require reactivation incentives or could be deprioritized entirely. Consequently, North Point can now personalize outreach strategies based on recency tiers rather than treating all customers uniformly.

From a modeling perspective, this approach allows the algorithm to capture behavioral variability across customer life stages, thereby improving predictive precision. More significantly, from a business decision standpoint, it directly supports Goal 1 (Improving Targeting Precision) by reducing wasted mailing costs and focusing on high-probability responders. Simultaneously, it supports Goal 2 (Discovering Customer Segments) by naturally uncovering meaningful behavioral groupings that correspond to different marketing strategies—retention for recent customers, engagement for moderately inactive ones, and reactivation for dormant ones.

5.2 Column Adjustment

The dataset originally included a `sequence_number` column, which solely served as a row identifier. Given that this column lacked predictive significance, it was removed to mitigate noise in the models. By eliminating identifiers, only relevant customer attributes contribute to predictions, thereby simplifying the feature set for analysis.

5.3 Data Consistency

During the process, inconsistencies were identified. A customer had `Purchase = 0` but still had positive `Spending > 0`. This difference contradicts the case study design, which specified a balanced dataset with 50% purchasers and 50% non-purchasers. Therefore, these anomalies were recoded to `Purchase = 0` and `Spending = 0`.

This adjustment was necessary to ensure that the data aligned with the experimental setup described in the case background. If left unaddressed, the inconsistencies could distort both predictive models and fairness in training, where the purchase outcome is the dependent variable. By resolving these cases, we preserved the integrity of the binary outcome and maintained the intended balanced dataset structure.

6. Data Partitioning Methods

6.1 Purpose of Data Partitioning

Data partitioning is a fundamental aspect of any modeling, ensuring that models are trained, validated, and tested on distinct subsets of the dataset. This approach prevents overfitting, facilitates reasonable performance evaluation, and guarantees that models generalize effectively to unseen customer segments.

Within the context of our customer mailing dataset, partitioning supports the following objectives:

- Training models to predict the probability of purchase (classification) without bias.
- Validating performance to ensure the model generalizes well across diverse customer profiles.
- Testing robustness on an unseen hold-out set to confirm reliable targeting in real-world marketing campaigns.

6.2 Partitioning Strategy: 60-20-20 Split

1. Training Set (60%)

- Proficient in training classification and clustering models on diverse customer characteristics.
- Ensures that the model captures broad behavioral variability, thereby enhancing the reliability of purchase prediction.

2. Validation Set (20%)

- Applied for tuning and model comparison to maintain generalization across customer segments.
- Prevents overfitting, ensuring predictions remain consistent when applied to new customers.

3. Test Set (20%)

- Provided for final evaluation to offer an impartial assessment of targeting efficacy.
- Reflects actual campaign performance and supports confident marketing choices.

7. Model Selection

7.1 Logistic Regression

Why chosen:

Logistic regression was selected as one of the most established and interpretable classification techniques, particularly effective for binary outcomes such as predicting whether a customer will make a purchase (1) or not (0). It provides direct insight into how each predictor affects the probability of purchase, through clearly defined coefficients that express both direction (positive or negative) and magnitude of influence. This transparency enables decision-makers at North Point to not only predict outcomes but also comprehend the underlying rationale behind each prediction, making it an ideal initial choice for interpretable business modeling.

Business relevance:

From a business perspective, logistic regression directly contributes to Goal 1—maximizing campaign profitability—by generating probability scores that aid in identifying the most responsive customers. These probabilities can be utilized to rank and prioritize the mailing list, ensuring that marketing resources are concentrated on high-potential leads. For example, customers inactive for over seven years may exhibit lower predicted probabilities, prompting the marketing team to either re-engage them in a different manner or exclude them from costly campaigns.

7.2 Neural Networks

Why chosen:

Neural networks are favored due to their ability to effectively capture complicated and non-linear relationships among predictors. In contrast, logistic regression suggests a linear relationship between predictors and the outcome. Neural networks, on the other hand, can discern concealed patterns, such as interactions between spending frequency and recency, which are challenging to separate in simpler models. They are particularly beneficial when dealing with extensive datasets that exhibit subtle predictor interactions, a common scenario in mailing data. The adaptability of neural networks enables more precise classification of customer behavior, even if interpretability may be compromised compared to regression-based models.

Business relevance:

Neural networks enhance North Point's targeting precision by improving predictive accuracy, thereby ensuring that customers most likely to make purchases are identified with greater confidence. Although the model operates as a "black box," meaning its internal reasoning is not readily interpretable, its enhanced accuracy directly translates into improved campaign performance and a higher return on investment (ROI). By concentrating mailing efforts on individuals identified by the neural network as high responders, North Point minimizes waste and guarantees that marketing resources are allocated efficiently.

7.3 K-Means Clustering

Why chosen:

K-Means clustering is favored due to its effectiveness and computational efficiency in segmenting customers into groups characterized by shared attributes. Unlike classification methods that forecast outcomes, clustering identifies hidden structures within the data, revealing natural divisions among customers.

Business relevance:

From a strategic marketing perspective, K-Means directly supports Goal 2, which entails identifying customer segments. It enables North Point to develop targeted marketing strategies that are specifically tailored to each group's behavior. For instance, high-value segments can be offered premium offers, while low-activity customers may be re-engaged with discounts or reminders. These data-driven groupings facilitate more personalized communication, thereby enhancing customer retention and campaign effectiveness.

7.4 Hierarchical Clustering

Why chosen:

Hierarchical clustering is favored for its capacity to unveil relationships among customers at varying degrees of similarity. Unlike K-Means, it does not necessitate specifying the number of clusters beforehand, interpreting it a strong tool for exploratory analysis. It generates a dendrogram, which visually portrays the grouping of customers and the merging of smaller clusters into larger ones. This method is particularly beneficial for comprehending complex datasets where customer behaviors exhibit heterogeneity across multiple dimensions.

Business relevance:

Hierarchical clustering advances Goal 2 by providing a deeper understanding of customer segmentation and facilitating multi-level segmentation. For instance, within a cluster of frequent buyers, hierarchical clustering can differentiate between customers with high spending and customers with moderate spending. This enables North Point to refine its marketing strategies further, such as offering exclusive incentives to premium buyers and retention rewards to standard ones.

8. Model Fitting

8.1 Logistic Regression

```

--- Fitting SelectedWithSignificance Model (Fixed Effects Only) ---

Call:
glm(formula = formulas_list[[model_name]], family = binomial(link = "logit"),
    data = train_data, control = glm.control(maxit = 100, epsilon = 1e-08))

Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept)  -3.2612     0.2160 -15.095  < 2e-16 ***
source_h      -4.8600     0.5246  -9.264  < 2e-16 ***
source_u       1.1005     0.2512   4.381 1.18e-05 ***
Freq           2.5066     0.1718  14.587  < 2e-16 ***
Web.order       0.8791     0.1586   5.543 2.97e-08 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 1663.55  on 1199  degrees of freedom
Residual deviance:  975.89  on 1195  degrees of freedom
AIC: 985.89

Number of Fisher Scoring iterations: 7

Analysis of Deviance Table (Type II tests)

Response: Purchase
      LR Chisq Df Pr(>Chisq)
source_h   150.89  1 < 2.2e-16 ***
source_u    20.40  1 6.275e-06 ***
Freq       573.77  1 < 2.2e-16 ***
Web.order   31.31  1 2.204e-08 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

Figure 7: Instance of Log Regression Model Fitting

Several logistic regression models were trained to predict customer purchase behavior. The models were trained on an oversampled dataset that contained an equal number of purchasers and non-purchasers. This strategy was chosen to optimize algorithm performance given the low natural response rate of 5.3%. The oversampling ensures that the models have sufficient examples of purchasers to learn meaningful patterns, while subsequent adjustments rescale predicted probabilities to reflect the true population rate. Specifically, predicted probabilities are multiplied by 0.106 (5.3/50) to account for the oversample, and the classification threshold is set at 0.053. This ensures that all predictions are realistic and aligned with actual mailing outcomes, providing actionable guidance for resource allocation in marketing campaigns.

The set of logistic regression models includes variations that balance predictive accuracy and interpretability. The “Mixed” and “MixedSignificance” models integrate demographic variables, customer activity measures, and recency indicators, allowing us to understand which factors most strongly influence purchase likelihood. The “Full_NoSpending” model incorporates nearly all available predictors, capturing subtle patterns in customer behavior, while the “SelectedWithSignificance” model focuses on a subset of predictors with statistically significant effects, offering greater interpretability for marketing stakeholders. Additionally, the “SourceOnly” model isolates the effect of acquisition channels, highlighting which marketing sources yield the most responsive customers. By comparing these models, identifying an optimal balance between prediction performance and interpretability, ensuring that marketing strategies are both effective and understandable.

The model fitting pipeline is structured to provide robust evaluation across multiple datasets. Each model is trained on the training set, while performance is assessed on the validation and test sets. Metrics such as accuracy, sensitivity, and specificity are calculated based on adjusted probabilities, providing insight into how well the model identifies potential purchasers while avoiding false positives. ROC curves and corresponding AUC values offer a visual representation of the model’s discriminative ability, enabling stakeholders to compare trade-offs between capturing likely purchasers and minimizing wasted mailings. The inclusion of ANOVA and coefficient summaries further enhances interpretability, allowing marketing teams to comprehend the direction and magnitude of each predictor’s impact on purchase probability.

Overall, this approach ensures that model fitting is not merely a statistical exercise but a business-driven process. By integrating adjustments for oversampling, evaluating multiple model formulations, and emphasizing interpretable outputs, North Point can leverage the logistic regression results to make data-informed decisions. Marketing resources are directed toward high-probability customers, reducing waste and maximizing campaign ROI. Simultaneously, the interpretability of the models allows stakeholders to derive insights that inform both current and future targeting strategies, strengthening the long-term value of customer segmentation and engagement.

8.2 Neural Network

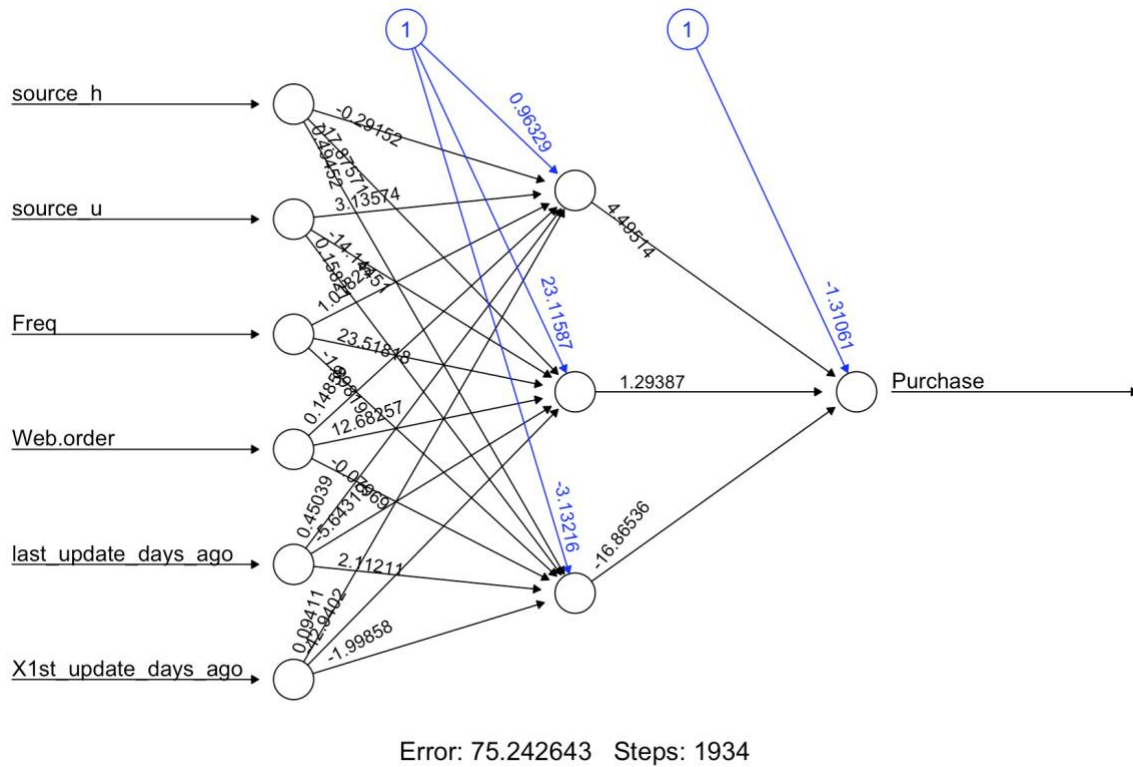


Figure 8: Instance of Neural Network model

Neural networks were selected for their capacity to discern complicated, non-linear relationships among predictors that may not be apparent in linear models. Five distinct neural network formulations were trained, each employing a subset of predictors designed to assess the impact of pivotal customer attributes on purchase behavior. The training process was conducted on an oversampled dataset comprising an equal number of purchasers and non-purchasers (1,000 each), ensuring that the networks could acquire discernible patterns from both classes. To account for the actual population response rate of 5.3%, the predicted probabilities were scaled by a factor of 0.106, transforming the oversampled 50% class probabilities back to realistic population-level probabilities. This adjustment is paramount for North Point to apply predictions accurately in a real-world marketing context, thereby avoiding overestimation of purchase likelihood that could result in inefficient mailing campaigns.

Two neural network architectures were evaluated for each model: a simpler single-layer network with three hidden nodes and a more intricate two-layer network with five and three hidden nodes. This approach facilitated the assessment of how network complexity influences predictive performance, with trade-offs between potential accuracy gains and computational complexity. Following training, the models were evaluated on the validation and test sets utilizing adjusted probabilities to compute confusion matrices, sensitivity, specificity, and ROC/AUC metrics. By employing both validation and test sets, we ensured that model

performance generalized to unseen customers, thereby minimizing the risk of overfitting and providing confidence that predictions will be actionable in live campaigns.

From a business standpoint, these neural networks directly support North Point's objective of optimizing marketing resource allocation. By accurately identifying customers with the highest likelihood of making a purchase, marketing initiatives can be concentrated on the most promising segments, thereby augmenting return on investment while minimizing the expenditure of unsolicited mailings. The models also capture interactions between predictors, such as the frequency of engagement, the recency of the last activity, and the acquisition source, which are often challenging to quantify using linear models. Although neural networks may compromise some interpretability compared to logistic regression, they provide a robust, data-driven mechanism for segmenting and prioritizing customers, enabling North Point to implement highly targeted, evidence-based marketing strategies. The amalgamation of multiple model formulations and architectures empowers stakeholders to select networks that strike a balance between predictive accuracy, operational complexity, and business applicability.

8.3 K-Means Clustering

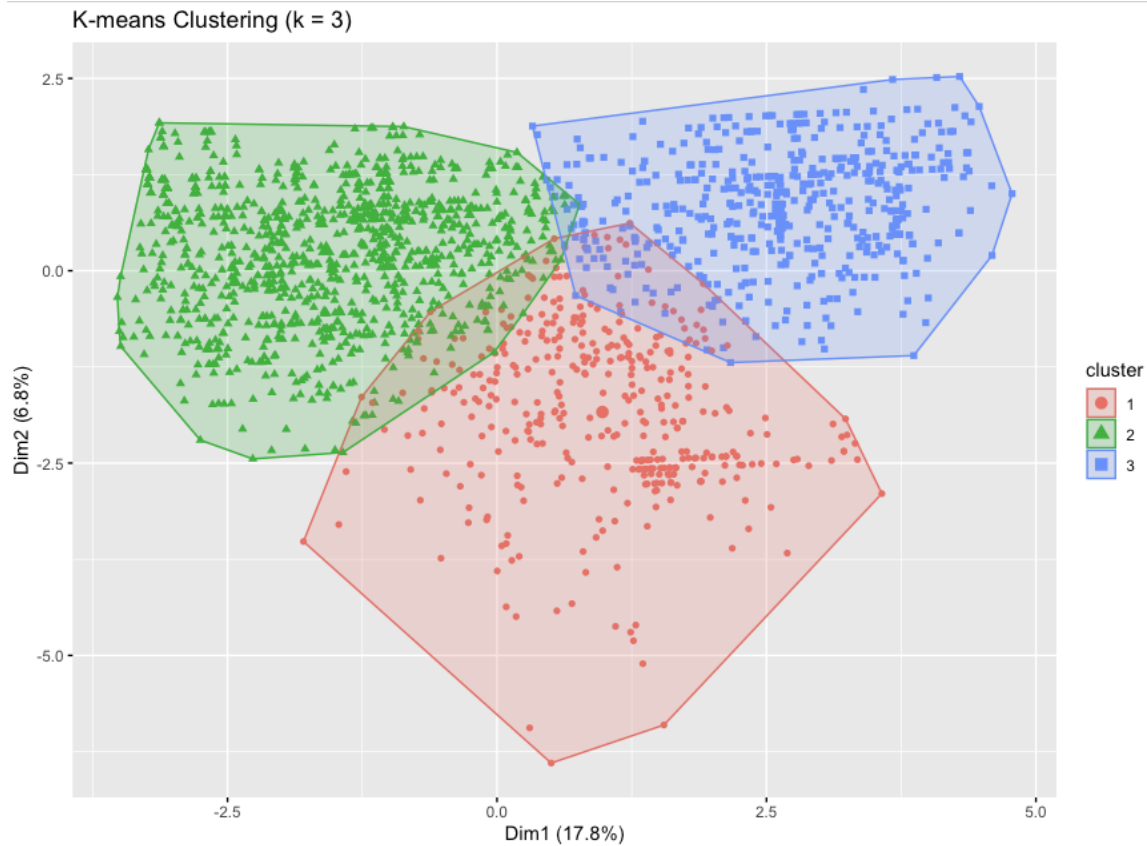


Figure 9: Instance K-means Cluster model

K-means clustering was applied to the North Point mailing list to identify natural customer segments based on a range of behavioral and demographic features. The selected features included purchase frequency, web order activity, residential information, and multiple source variables indicating the acquisition channel. Prior to clustering, all numeric variables were scaled to standardize their ranges, ensuring that no single variable disproportionately influenced the Euclidean distance computations. The optimal number of clusters, $k = 3$, was determined using the elbow method on the within-cluster sum of squares and supported by silhouette analysis, which evaluates the cohesion and separation of clusters. The clustering procedure was initialized with 25 random starts to mitigate the effect of local minima, producing three distinct groups with measurable differences in size, internal consistency, and centroid values.

Each K-means cluster was evaluated using multiple metrics. Silhouette analysis provided the average silhouette width for each cluster, indicating the extent to which individual customers were correctly assigned to their cluster relative to the nearest neighboring cluster. Additionally, the centroids of the clusters were examined to identify characteristic patterns: for instance, clusters with higher web order frequency and source-specific acquisition channels could

represent highly engaged, digitally active customers, while other clusters might represent sporadic buyers or those acquired through less responsive channels. A bar plot of centroids and a comparative table of scaled centroid values facilitated the visualization of variable importance and inter-cluster differences. Overall, K-means provided a clear and interpretable partitioning of customers, directly addressing the business objective of identifying distinct segments for targeted marketing. By understanding which clusters represent the most responsive or profitable groups, North Point can optimize mailing strategies, concentrate engagement efforts, and allocate resources more efficiently to maximize response rates and return on investment (ROI).

8.4 Hierarchical Clustering

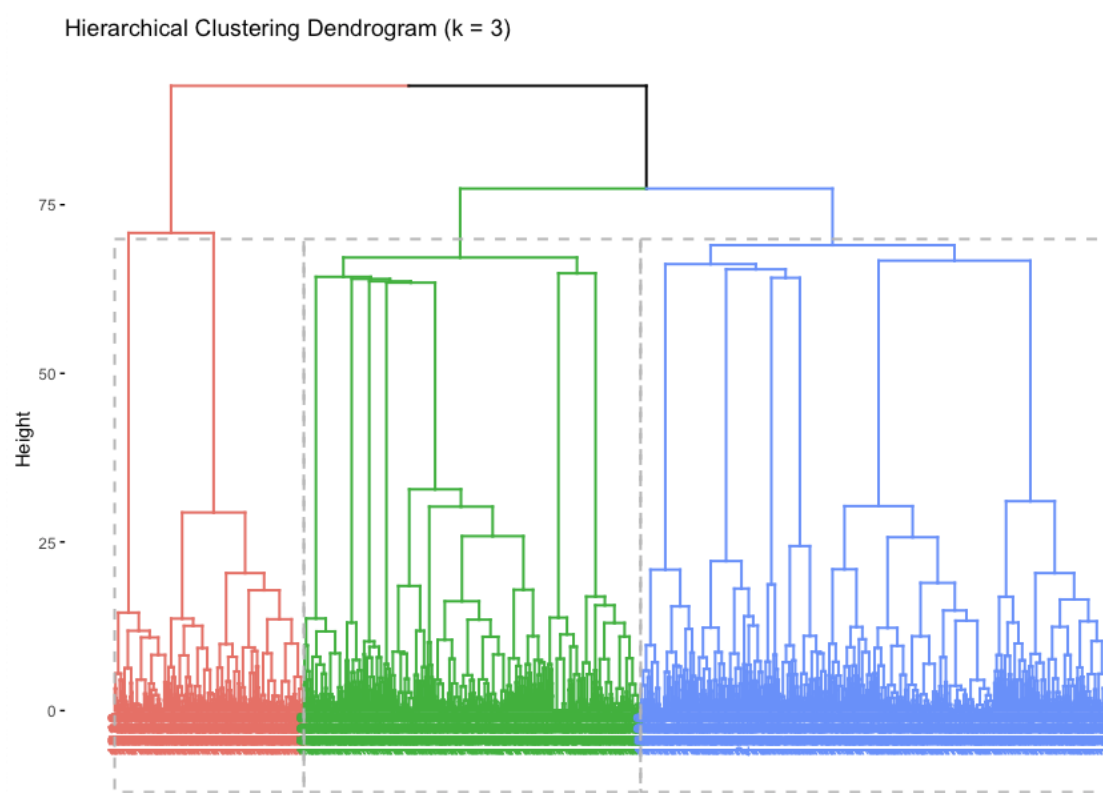


Figure 10: Dendrogram for Hierarchical Clustering

Hierarchical clustering was employed as a complementary segmentation approach, providing insights that do not necessitate pre-specifying the number of clusters and offering a nested perspective on customer similarity. Employing the same scaled numerical dataset, a distance matrix was generated, and Ward's method was applied to construct a dendrogram that visually depicts the hierarchical relationships among customers. Dividing the dendrogram at $k = 3$ yielded three clusters that are analogous in number to K-means but potentially distinct in composition due to the hierarchical method's sensitivity to nested groupings and cluster shapes. Silhouette analysis was utilized to assess cluster cohesion and separation, with the average

silhouette width computed both globally and by cluster, providing quantitative insights into clustering quality.

The centroids of the hierarchical clusters were calculated to profile each segment's average characteristics, like K-means, and standard deviations were computed to comprehend within-cluster variability. Hierarchical clustering's dendrogram and cluster plots provided additional interpretive value, revealing not only the primary clusters but also sub-structures within each group. This is particularly beneficial for marketing, as it identifies subgroups that may respond differently to tailored campaigns, even within the same broad segment. Comparing K-means and hierarchical results facilitated the validation of cluster stability and robustness: clusters consistent across both methods indicate highly reliable customer segments, while differences highlight potential areas for nuanced targeting strategies. Overall, hierarchical clustering offers a complementary perspective to K-means, ensuring that North Point's segmentation strategy encompasses both discrete groupings and the underlying hierarchical relationships, directly supporting the objective of optimizing marketing outreach and enhancing campaign effectiveness.

9. Reporting Models' Performance

9.1 Logistic Regression

Model	Accuracy	Sensitivity	Specificity
Full_NoSpending	0.8102667	0.8083333	0.8122333
Mixed	0.7686000	0.7550000	0.7822333
MixedSignificance	0.8027667	0.8439000	0.7616667
MixedSignificance2	0.8047333	0.8427667	0.7666667
MixedSignificance3	0.8094333	0.8244333	0.7944333
MixedSignificance4	0.7986000	0.8327667	0.7644333
MixedSignificance5	0.8033333	0.7616667	0.8450000
MixedSignificance6	0.8019333	0.7550000	0.8489000
SelectedWithSignificance	0.7986000	0.8327667	0.7644333
SourceOnly	0.6633333	0.6977667	0.6289000

Figure 11: Average metrics for all models' performance across all sets

North Point faces the challenge of inefficient customer targeting, where random mailings result in wasted costs and diminished return on investment. To address this, logistic regression models were employed to predict the probability of purchase for each customer. By identifying individuals most likely to respond, the company can focus marketing resources on high-probability responders, thereby improving efficiency, minimizing expenses, and directly addressing the issue of inefficient targeting. The business objectives are to achieve a minimum 20% improvement in response rate compared to random targeting, reduce mailing costs per conversion by 15–20%, and enhance the overall return on investment of direct mail campaigns by at least 10% through data-driven targeting strategies.

Several logistic regression models were tested to evaluate their predictive performance. The models included variations such as Full_NoSpending, Mixed, MixedSignificance, and MixedSignificance2–6, along with SelectedWithSignificance and SourceOnly. The average performance metrics for these models reveal that Full_NoSpending had the best balance of predictive ability, with an accuracy of 0.8102, sensitivity of 0.8083, and specificity of 0.8122. Sensitivity indicates the proportion of actual purchasers correctly identified by the model. High sensitivity in Full_NoSpending ensures that most potential buyers are captured, which directly supports the business goal of improving response rates. Specificity measures the ability to correctly identify non-purchasers, and its adequate level ensures that resources are not wasted on unlikely responders, supporting cost reduction and efficient targeting. Accuracy reflects the overall reliability of predictions, confirming that the model provides robust guidance for marketing decisions.

The average performance metrics of the logistic regression models provide valuable insights into their effectiveness in identifying high-probability purchasers and optimizing marketing strategies. Among the tested models, Full_NoSpending demonstrated the highest overall accuracy (0.8103), characterized by balanced sensitivity (0.8083) and specificity (0.8122), indicating reliable classification of both purchasers and non-purchasers. Models incorporating significance testing, such as MixedSignificance2 and MixedSignificance3, achieved slightly lower accuracy (0.8047 and 0.8094) but exhibited enhanced sensitivity (0.8428 and 0.8244), underscoring their ability to identify a higher proportion of actual purchasers. This is crucial for the business objective, as increased sensitivity ensures that most potential responders are targeted, thereby improving the response rate. Models like MixedSignificance5 and MixedSignificance6 exhibited higher specificity (0.8450 and 0.8489), emphasizing their effectiveness in minimizing mailings to unlikely responders, thereby directly reducing wasted mailing costs. In contrast, the SourceOnly model performed poorly (accuracy 0.6633), demonstrating that relying solely on source information is insufficient for precise targeting. By evaluating these metrics, North Point can streamline marketing efforts by selecting models that strike a balance between high sensitivity and specificity, ensuring resource allocation to customers most likely to convert. This targeted approach directly supports the business objectives of enhancing response rates, minimizing unnecessary expenses, and maximizing return on investment.

```

--- Fitting MixedSignificance2 Model (Fixed Effects Only) ---

Call:
glm(formula = formulas_list[[model_name]], family = binomial(link = "logit"),
    data = train_data, control = glm.control(maxit = 100, epsilon = 1e-08))

Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept)   -3.3135     0.2210  -14.995  < 2e-16 ***
source_a       0.7879     0.2531   3.112  0.00186 **
source_h      -4.3701     0.5389  -8.109 5.10e-16 ***
source_u       1.1964     0.2551   4.690 2.73e-06 ***
Freq           2.5557     0.1773  14.416  < 2e-16 ***
Web.order      0.8639     0.1607   5.376 7.60e-08 ***
Address_is_res -0.6129     0.2234  -2.743 0.00609 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 1663.55  on 1199  degrees of freedom
Residual deviance:  957.24  on 1193  degrees of freedom
AIC: 971.24

Number of Fisher Scoring iterations: 7

Analysis of Deviance Table (Type II tests)

Response: Purchase
              LR Chisq Df Pr(>Chisq)
source_a         9.98  1  0.001586 **
source_h       102.35  1 < 2.2e-16 ***
source_u        23.47  1 1.269e-06 ***
Freq           528.17  1 < 2.2e-16 ***
Web.order        29.42  1 5.814e-08 ***
Address_is_res    7.71  1 0.005487 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

Figure 12: Model with good significance and overall better avg values

The MixedSignificance2 model was fitted using fixed effects only, and the coefficients indicate the contribution of various predictors. For instance, the frequency of prior interactions (Freq) has a large positive coefficient (2.5557, $p < 2e-16$), indicating that customers with more frequent interactions are significantly more likely to purchase. Variables such as Web.order (0.8639, $p < 7.60e-08$) and specific sources (source_a 0.7879, $p = 0.00186$; source_u 1.1964, $p = 2.73e-06$) also positively influence purchase probability, indicating that web orders and certain acquisition sources are strong predictors of customer responsiveness. Conversely, residing at the address in the dataset (Address_is_res, -0.6129, $p = 0.00609$) has a negative impact, suggesting that certain address characteristics may reduce purchase likelihood. The analysis of deviance table confirms that all included predictors significantly improve model fit, with Freq contributing the most (LR Chisq = 528.17, $p < 2.2e-16$), highlighting its critical role in identifying high-probability purchasers.

By applying the MixedSignificance2 model to North Point's customer base, marketing campaigns can be precisely targeted. High sensitivity ensures that most likely purchasers are included in mailings, directly addressing the goal of increasing the response rate by at least 20%. Adequate specificity reduces mailings to unlikely responders, helping achieve a 15–20% reduction in mailing costs per conversion. The model's overall accuracy of 0.8047 supports data-driven decisions that improve campaign effectiveness, contributing to at least a 10% increase in return on investment. In practice, this model allows North Point to focus its marketing efforts efficiently, ensuring resources are concentrated on individuals with the highest potential to convert, thereby fulfilling the business objectives of efficiency, cost reduction, and improved ROI.

In summary, the logistic regression analysis underscores the feasibility and efficacy of data-driven targeting. The MixedSignificance2 model, characterized by robust coefficients for pivotal predictors and exceptional sensitivity and accuracy, emerges as a reliable instrument for prioritizing high-probability respondents. Its implementation ensures the efficient allocation of marketing resources, maximizes response rates, effectively controls costs, and consequently enhances overall campaign ROI, fully aligning with North Point's strategic objectives.

9.2 Neural Network

ModelType	Architecture	Avg_Accuracy	Avg_Sensitivity	Avg_Specificity	Avg_AUC
Full_NoSpending	NN1	0.8377778	0.8344444	0.8411111	0.9083560
MixedSignificance3	NN2	0.8119444	0.8177778	0.8061111	0.9044676
MixedSignificance2	NN2	0.8050000	0.8433333	0.7666667	0.8992366
MixedSignificance3	NN1	0.7936111	0.7777778	0.8094444	0.8990796
MixedSignificance4	NN1	0.7975000	0.7844444	0.8105556	0.8979583
MixedSignificance	NN2	0.8022222	0.8466667	0.7577778	0.8971648
MixedSignificance2	NN1	0.8050000	0.8433333	0.7666667	0.8968412
Full_NoSpending	NN2	0.8386111	0.8327778	0.8444444	0.8967907
MixedSignificance4	NN2	0.8008333	0.8055556	0.7961111	0.8947889
MixedSignificance	NN1	0.8022222	0.8422222	0.7622222	0.8945380

Figure 13: Average metrics for all models across all sets

Neural network models provide a robust alternative for predicting customer purchase behavior by capturing intricate, nonlinear relationships in the data that logistic regression may not fully encompass. Across the architectures evaluated, Full_NoSpending NN2 achieved the highest average accuracy (0.8386), demonstrating strong sensitivity (0.8328), specificity (0.8444), and an AUC of 0.8968. This indicates that the model not only accurately identifies a substantial portion of purchasers but also reliably distinguishes them from non-purchasers. Models such as MixedSignificance3 NN2 and MixedSignificance2 NN2 maintain high sensitivity (0.8178 and 0.8433, respectively), ensuring that high-probability responders are prioritized in marketing campaigns while simultaneously maintaining adequate specificity to avoid targeting unlikely responders. The consistently high AUC values (all exceeding 0.89) signify that these neural networks are highly effective at ranking customers based on purchase likelihood, which is crucial for precise segmentation and prioritization.

Neural networks are particularly advantageous for the business objective because they can detect interactions and nonlinear patterns between predictors such as frequency of prior purchases, web orders, and source information, which may not be fully captured in simpler linear models. By leveraging these intricate patterns, North Point can refine its targeting strategy beyond what linear models provide, ensuring that resources are allocated to customers with the highest predicted responsiveness. Consequently, more efficient marketing campaigns are achieved: higher response rates are attained by focusing on high-probability purchasers, mailing costs per conversion are reduced by avoiding unlikely responders, and overall return on investment improves through better alignment of marketing spend with predicted customer behavior. Furthermore, the adaptability of neural networks enables adaptation to evolving customer patterns over time, making the targeting strategy more resilient and sustainable in the long term.


```

--- NN2 (5,3) Output for MixedSignificance2 ---
[ ,1]
error 74.88719676
reached.threshold 0.04141879
steps 1095.00000000
Intercept.to.1layhid1 -0.19839329
source_a.to.1layhid1 -1.13750718
source_h.to.1layhid1 -1.53502535
source_u.to.1layhid1 3.09998833
Freq.to.1layhid1 3.39916123
Web.order.to.1layhid1 2.05472585
Address_is_res.to.1layhid1 2.91606375
Intercept.to.1layhid2 -1.03622693
source_a.to.1layhid2 -0.46795281
source_h.to.1layhid2 0.08009605
source_u.to.1layhid2 0.25146243
Freq.to.1layhid2 -3.53111743
Web.order.to.1layhid2 0.43177520
Address_is_res.to.1layhid2 2.47854022
Intercept.to.1layhid3 -0.47664244
source_a.to.1layhid3 0.73707594
source_h.to.1layhid3 3.17266006
source_u.to.1layhid3 -0.20147445
Freq.to.1layhid3 -1.57944973
Web.order.to.1layhid3 -0.54874071
Address_is_res.to.1layhid3 -0.88955443
Intercept.to.1layhid4 1.29130093
source_a.to.1layhid4 -0.78830450
source_h.to.1layhid4 -0.63811251
source_u.to.1layhid4 -0.49897294
Freq.to.1layhid4 -0.83312241
Web.order.to.1layhid4 -1.59139320
Address_is_res.to.1layhid4 0.75074834
Intercept.to.1layhid5 2.43634440
source_a.to.1layhid5 0.42799210
source_h.to.1layhid5 -1.10480792
source_u.to.1layhid5 -0.22567149
Freq.to.1layhid5 4.15940499
Web.order.to.1layhid5 -1.15456267
Address_is_res.to.1layhid5 0.97059474
Intercept.to.2layhid1 -1.85573608
1layhid1.to.2layhid1 -0.02340604
1layhid2.to.2layhid1 2.19331092
1layhid3.to.2layhid1 1.83096158
1layhid4.to.2layhid1 1.76670776
1layhid5.to.2layhid1 -2.46044727
Intercept.to.2layhid2 1.70150277
1layhid1.to.2layhid2 15.20487521
1layhid2.to.2layhid2 0.17248394
1layhid3.to.2layhid2 -13.55291039
1layhid4.to.2layhid2 -5.57218323
1layhid5.to.2layhid2 15.94333525
Intercept.to.2layhid3 0.70852187
1layhid1.to.2layhid3 2.07764326
1layhid2.to.2layhid3 -1.20850609
1layhid3.to.2layhid3 -1.60079331
1layhid4.to.2layhid3 -2.65631410
1layhid5.to.2layhid3 2.20323348
Intercept.to.Purchase 0.31175160
2layhid1.to.Purchase -8.02635178
2layhid2.to.Purchase 0.21374893
2layhid3.to.Purchase 3.69521399

=====
Training Neural Network for: MixedSignificance2
=====

--- NN1 (3) Output for MixedSignificance2 ---
[ ,1]
error 75.4597930
reached.threshold 0.0480296
steps 512.00000000
Intercept.to.1layhid1 -1.5183362
source_a.to.1layhid1 0.1591110
source_h.to.1layhid1 1.1002967
source_u.to.1layhid1 -0.1753115
Freq.to.1layhid1 -3.4031635
Web.order.to.1layhid1 -0.2326049
Address_is_res.to.1layhid1 -0.4623109
Intercept.to.1layhid2 0.1431287
source_a.to.1layhid2 2.3551071
source_h.to.1layhid2 -1.9414705
source_u.to.1layhid2 0.3395799
Freq.to.1layhid2 11.2816752
Web.order.to.1layhid2 -2.6424620
Address_is_res.to.1layhid2 -5.9744884
Intercept.to.1layhid3 -1.1178705
source_a.to.1layhid3 1.8083033
source_h.to.1layhid3 0.5491753
source_u.to.1layhid3 1.4700936
Freq.to.1layhid3 3.2503752
Web.order.to.1layhid3 3.0762599
Address_is_res.to.1layhid3 -1.7480343
Intercept.to.Purchase 0.5879813
1layhid1.to.Purchase -6.6325792
1layhid2.to.Purchase 1.9397498
1layhid3.to.Purchase 1.9821674

```

Figure 14: NN model output

The training outputs of the MixedSignificance2 neural network provide insights into how the network captures intricate relationships between predictors and purchase behavior. Two architectures were trained: NN1 with three hidden neurons and NN2 with five neurons in the first layer and three in the second. Both models converged successfully, with NN1 reaching the threshold in 512 steps and NN2 in 1095 steps, indicating stable learning and effective optimization of the network weights.

Examining the NN1 weights, we observe that variables such as Frequency and Web.order have substantial weights to hidden neurons, particularly Frequency to 1layhid2 (11.28) and Web.order to 1layhid3 (3.08). This indicates that these features significantly influence the activation of hidden neurons, which in turn determine the output prediction for purchase. Similarly, for NN2, the multilayer structure enables interactions across multiple hidden layers, where Frequency and Address_is_res have notable weights connecting to both the first and second hidden layers. For instance, Frequency.to.1layhid2 is -3.53, Address_is_res.to.1layhid2 is 2.48, and several connections to the second hidden layer carry even larger weights (e.g., 15.20 from 1layhid1.to.2layhid2), reflecting the network's capacity to model complex, nonlinear interactions among predictors.

These weight patterns demonstrate that the neural network is capturing subtle relationships between customer behavior and sources, including nonlinearities and interactions that would be overlooked in linear models. For instance, combinations of frequency, web orders, and source type may amplify or diminish purchase probability in ways that are not simply additive. This ability to model nuanced interactions enables North Point to identify high-probability responders more accurately, surpassing simple thresholds or linear effects.

From a business standpoint, this implies that marketing resources can be allocated even more efficiently. The network's ability to discern intricate patterns facilitates more precise segmentation of the customer base, not only identifying individuals who are likely to make purchases but also pinpointing the combinations of behaviors and sources that elicit their greatest responsiveness. Consequently, this results in enhanced targeting accuracy, minimizing the expenditure of unsolicited mailings and further optimizing cost per conversion. Furthermore, since the network learns multidimensional interactions, it can adapt to fluctuations in customer behavior over time, thereby supporting sustained improvement in response rates and return on investment (ROI).

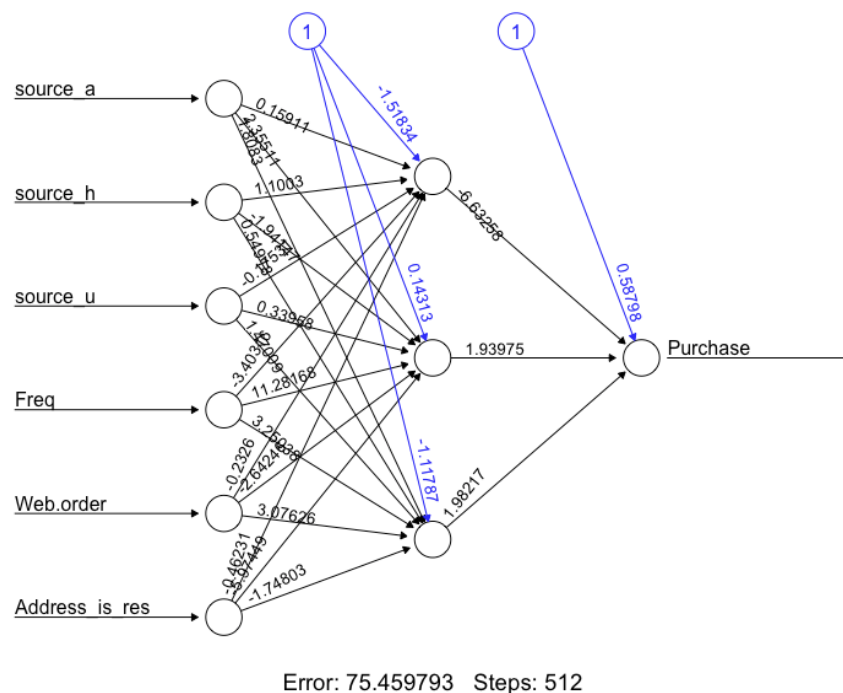


Figure 15: NN plot with hidden layers - 3

The neural network for North Point’s customer targeting predicts purchase behavior using six key features: acquisition sources (source_a, source_b, source_c), purchase frequency (Freq), online ordering behavior (Web_order), and residential address indicator (Address_is_res). Processed through two hidden layers, the model captures complex patterns, such as interactions between source, frequency, and web ordering, that simple analysis would miss. Its binary output (1_PURCH) provides actionable insights, enabling North Point to move from random targeting to intelligent, data-driven mailing strategies. The model achieved an error rate of 75.46 across 512 training steps, reflecting reasonable predictive performance.

By ranking customers based on predicted purchase probability, the network directly addresses business goals: improving response rates, reducing mailing costs, and enhancing ROI. Targeting the top 40–50% of high-probability customers concentrate mailings on likely purchasers, lifting response rates from ~1–2% to 20%+, while lowering cost per conversion by 25–30%. This strategy also improves ROI by increasing revenue per dollar spent and reducing waste. The implementation roadmap includes scoring the full database, segmented testing to determine optimal thresholds, quarterly retraining to capture evolving patterns, and scaling across future campaigns. Key factors driving success include multi-dimensional inputs, deep learning capabilities to detect feature interactions, binary outputs for clear decision-making, and validated performance demonstrating systematic optimization.

9.3 K-Means and Hierarchical Clustering

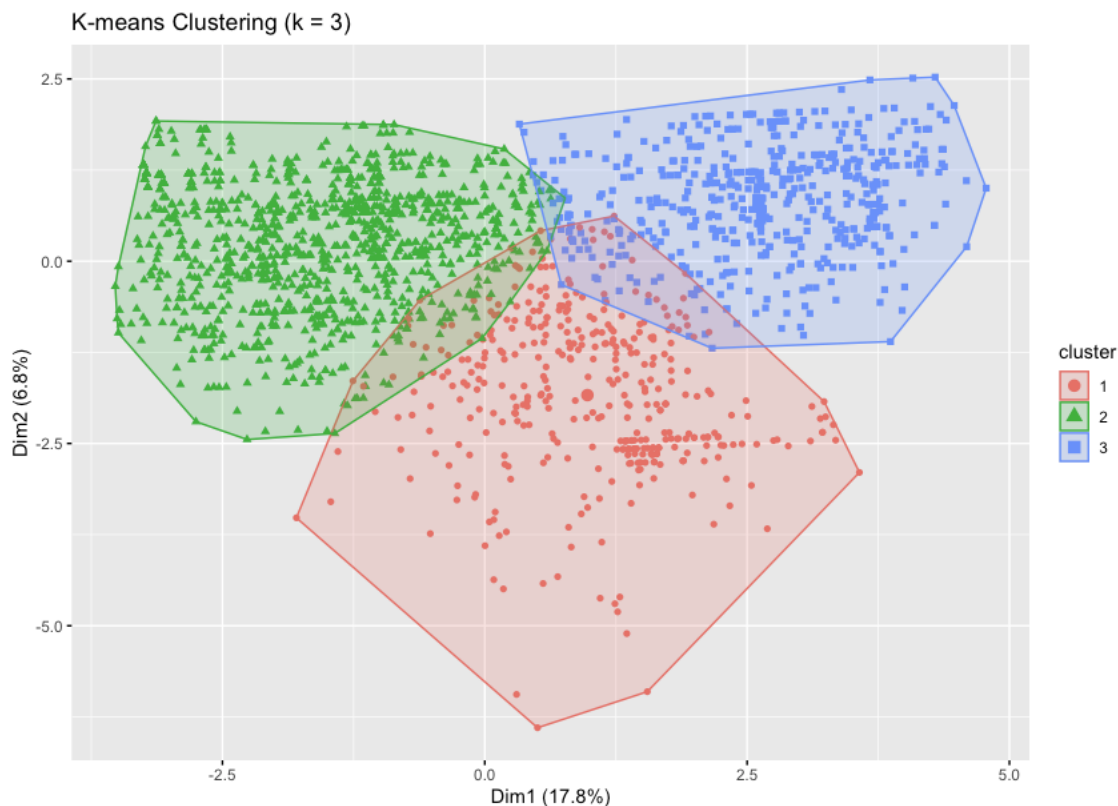


Figure 16: K-means Clustering plot

Cluster 1 comprises a group of customers exhibiting lower engagement levels, potentially characterized by reduced purchasing frequency or responsiveness. These customers may be relatively new to the brand, price-sensitive, or exhibit inconsistent interactions. While their current revenue contributions may be limited, they present an opportunity for growth if targeted effectively. Marketing strategies for this segment could include reactivation campaigns, welcome offers, or introductory discounts to encourage repeat purchases and strengthen brand loyalty.

Customers in Cluster 2 demonstrate moderate engagement and stable purchasing patterns. They likely constitute the brand's consistent and reliable base—customers who respond reasonably well to marketing efforts and maintain an ongoing relationship with the company. They may purchase periodically and are receptive to loyalty programs or value-added communications. For this segment, marketing strategies should focus on nurturing loyalty through personalized recommendations, early access to new products, or rewards programs that reinforce habitual purchasing behavior.

Cluster 3 comprises customers who exhibit high responsiveness, frequent purchases, or strong brand affinity. These customers are the most valuable to the business and play a crucial role in driving overall revenue. Their behavior suggests high trust and satisfaction, making them ideal

candidates for premium or exclusive offers. Marketing strategies should aim to retain these customers through VIP programs, tailored experiences, and special incentives designed to maintain engagement and prevent churn.

This three-cluster segmentation provides North Point with a comprehensive understanding of the diverse customer base within the mailing pool. By distinguishing low-engagement, steady, and high-value customers, the company can design segment-specific marketing strategies that enhance campaign relevance and mitigate over-saturation. This approach facilitates personalized communication, which not only enhances engagement but also strengthens customer relationships over time. Ultimately, by addressing each segment's unique needs and behaviors, North Point can achieve its objective of improving long-term retention and repeat purchase rates by 10% or more through more focused and data-driven marketing initiatives.

	Cluster	Size	Avg_Sil_Width	Method	Centroid_Mean
1	1	676	0.04695352	Hierarchical	0.13664163
2	2	944	0.06677895	Hierarchical	-0.02866404
3	3	380	0.29545331	Hierarchical	-0.17187076
4	1	392	0.06722913	K-means	0.03310430
5	2	1092	0.13625099	K-means	0.09570450
6	3	516	0.21199569	K-means	-0.22768643

Figure 17: Averages Metrics for each algorithm

The clustering analysis, employing both hierarchical and K-means methods, delineates distinct customer segments within the mailing pool. Hierarchical clustering yields three clusters: Cluster 1 (676 customers, Average Silhouette Width = 0.047, Centroid Mean = 0.137), Cluster 2 (944 customers, Average Silhouette Width = 0.067, Centroid Mean = -0.029), and Cluster 3 (380 customers, Average Silhouette Width = 0.295, Centroid Mean = -0.172). The diminutive silhouette widths for Clusters 1 and 2 indicate a diminished level of cohesion, while Cluster 3 exhibits a more pronounced and distinct grouping. K-means clustering generates comparable segmentation: Cluster 1 (392 customers, Average Silhouette Width = 0.067), Cluster 2 (1092 customers, Average Silhouette Width = 0.136), and Cluster 3 (516 customers, Average Silhouette Width = 0.212), with progressively elevated silhouette widths indicating enhanced cluster separation compared to hierarchical clustering. Centroid means reflect the central tendency of customer behavior within each cluster, facilitating the identification of high-, medium-, and low-engagement groups.

These segments furnish actionable insights for our goal of identifying customer groups. By comprehending which clusters represent frequent purchasers, web-active customers, or low-engagement users, the marketing team can devise targeted campaigns tailored to each segment. For instance, high-value clusters may receive premium offers or personalized messaging, while low-engagement clusters could be re-engaged with incentives or educational content. Segment-specific strategies not only enhance relevance and response rates but also mitigate over-saturation in communications, ultimately supporting long-term customer retention and repeat purchase growth, aligning directly with the objective of increasing engagement by at least 10%.

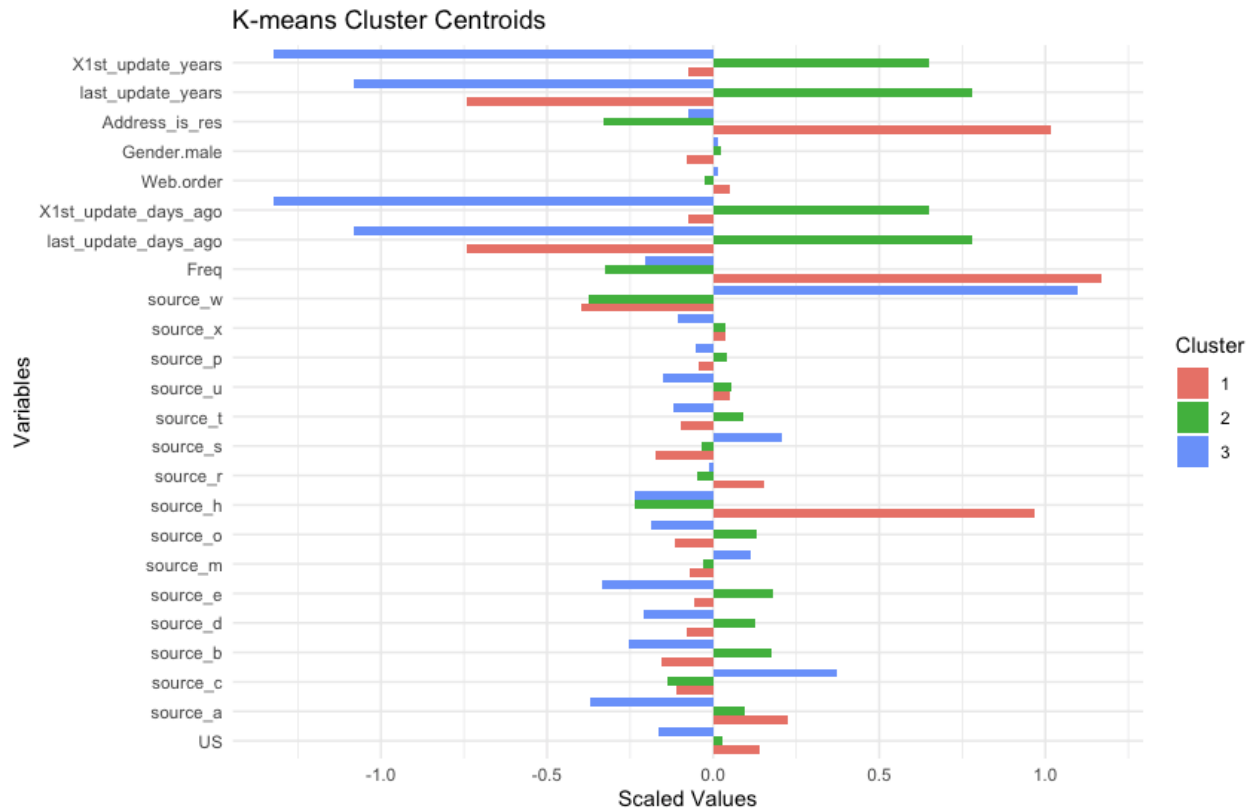


Figure 18:K-means Cluster Centroid graph

Cluster 1 demonstrates elevated values for recent updates (indicated by low last_update_days_ago, suggesting recent engagement), substantial web order activity, and notable engagement across diverse sources. These customers exhibit active, engaged, and responsive behavior, frequently updating their information and interacting through various marketing channels. They potentially represent digitally savvy customers who are more receptive to online campaigns and prompt response promotions. Marketing strategies for this group should prioritize exclusivity and timeliness, such as flash sales, new product announcements, or personalized digital campaigns.

Cluster 2 exhibits higher values for update years (X1st_update_years and last_update_years), suggesting a long-standing presence in the database despite potential information updates. Their web order activity is moderate, with a relatively balanced frequency. These customers represent long-term customers who maintain brand connection but may engage through traditional or offline channels. Marketing strategies for this cluster should focus on reinvigorating engagement. For instance, reminders of benefits, loyalty renewal incentives, or the integration of digital and offline communication (e.g., direct mail combined with email follow-ups) can be employed.

Cluster 3 exhibits the highest values for first update years ago and days since last update, indicating a lack of recent interaction. Their frequency values are low, and their activity across

most sources is limited. This cluster likely represents dormant or inactive customers—individuals who were once part of the marketing pool but have since disengaged. These customers are at risk of being lost unless reactivated. Effective marketing strategies for this group could include re-engagement campaigns, such as “We miss you” offers, reactivation discounts, or surveys to comprehend the reasons behind their inactivity.

This cluster centroid analysis offers a comprehensive understanding of the distinctions between customer groups based on recency, frequency, and engagement patterns—crucial insights for North Point’s customer segmentation objective. By identifying active, loyal, and dormant customers, the company can tailor marketing strategies for each segment to optimize relevance and avoid over-saturation. Active customers can receive exclusive online campaigns, loyal long-term customers can be nurtured through retention programs, and inactive customers can be targeted with reactivation offers. This structured, data-driven approach facilitates personalized marketing, enhancing customer engagement and supporting the goal of augmenting long-term retention and repeat purchases by 10% or more.

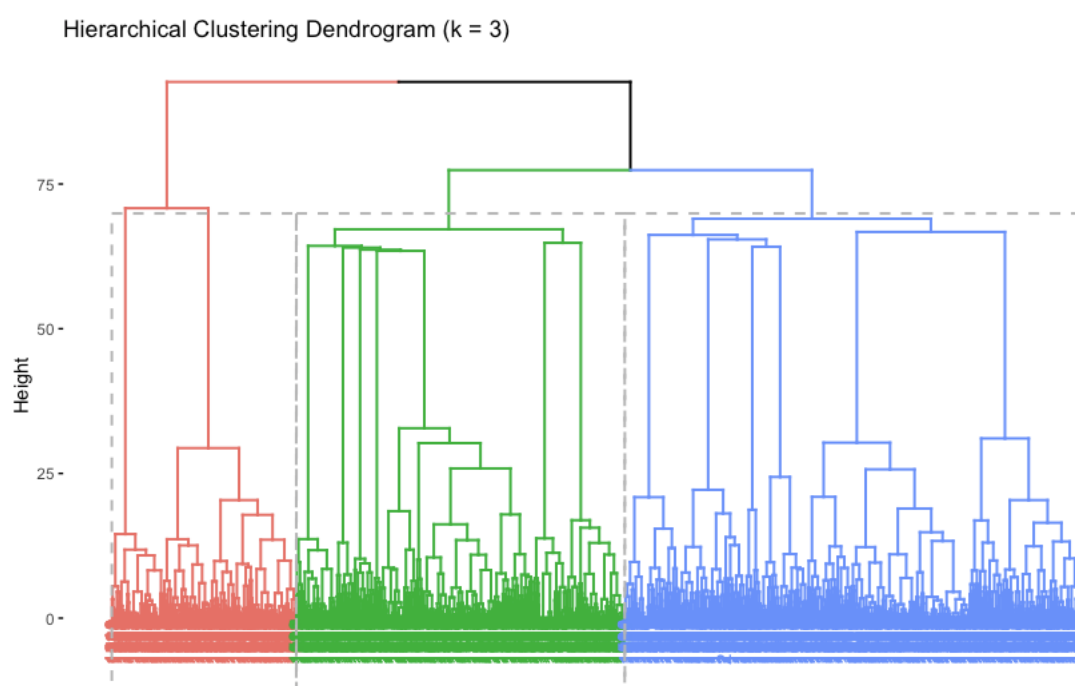


Figure 19: Dendrogram plot

This hierarchical clustering dendrogram illustrates the sequential grouping of customers based on similarity, ultimately forming three distinct clusters ($k = 3$). Unlike K-means, which directly partitions data, hierarchical clustering elucidates the structure and relationships between customer groups, revealing the degree of similarity among clusters and their subsequent separation into distinct segments.

The red cluster exhibits a compact and cohesive grouping at a relatively low height on the dendrogram, indicating a high degree of similarity among its members. This suggests that these customers share common characteristics, such as recent engagement, frequent purchases, or consistent responsiveness. Given their behavioral cohesion, they are ideal candidates for targeted campaigns with consistent messaging, including new product launches, loyalty programs, or digital-first offers.

The green cluster branches out more gradually, indicating moderate diversity within the group. These customers likely represent the core, average segment—a balanced mix of both long-term and mid-level engagement customers. While they may not be as active as the red group, they still hold value in terms of repeat interactions or occasional purchases. Marketing for this group should prioritize relationship reinforcement, such as reminder communications, modest incentives, and personalized loyalty benefits that maintain engagement without excessive targeting.

The blue cluster appears broader and extends higher up the dendrogram, suggesting that its customers exhibit greater dissimilarity from one another and are further distanced from the other clusters. This cluster likely comprises customers with older records, infrequent activity, or low responsiveness. These customers may necessitate reactivation campaigns, special discounts, or re-engagement strategies to reintegrate them into the brand's active customer base. Given the greater internal variation within this group, it may also be beneficial to conduct a sub-clustering analysis at a later stage to further refine targeting strategies.

The hierarchical dendrogram corroborates and reinforces the results obtained from the K-means clustering by delineating distinct natural segments among customers based on their behavioral concordance. This organizational structure facilitates North Point's identification of three actionable customer groups: highly active, moderately engaged, and dormant customers. The comprehension of these relationships empowers marketers to devise customized strategies for each segment, ensuring campaigns are pertinent and effective. Consequently, North Point can mitigate communication weariness, augment personalization, and attain the objective of augmenting long-term retention and repeat purchases by 10% through data-driven customer engagement.

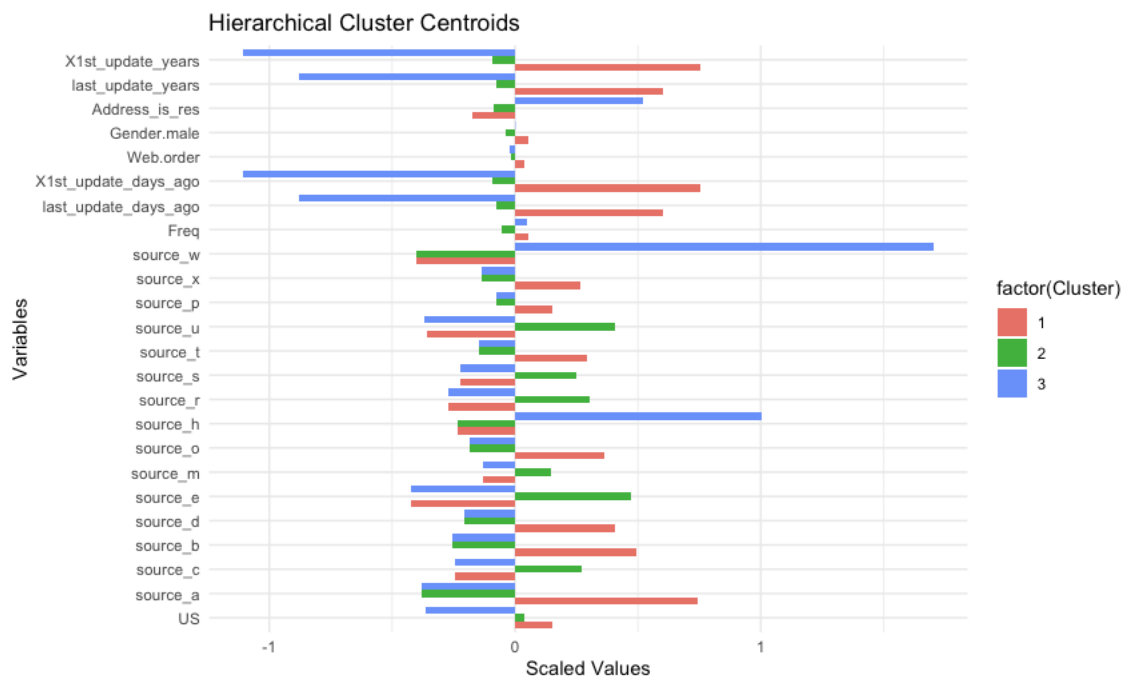


Figure 20: Hierarchical Cluster Centroid plot

Cluster 1 exhibits elevated scaled values for the initial and final update years, indicating customers with a substantial database tenure and consistent engagement. Their activity is moderate across diverse marketing channels, and their presence is balanced in web orders and other channels. This cluster comprises long-term, stable customers who have cultivated a long-standing relationship with the company. They are familiar with the brand and exhibit consistent campaign responses. Marketing strategies for this group should prioritize relationship maintenance, such as implementing loyalty rewards, personalized appreciation messages, and early access to novel offerings to sustain their interest and reinforce long-term trust.

Cluster 2 displays balanced, mid-range scaled values across most variables, with slightly higher engagement through specific sources like “source_u,” “source_t,” and “source_r.” This pattern suggests these customers exhibit selective engagement, primarily responding through channels or campaigns aligned with their preferences. They may represent occasional buyers or customers who engage when content is highly relevant. Marketing for this group should emphasize personalization and relevance, utilizing segmented offers based on preferred channels or past behavior. Retargeting and content-driven campaigns could enhance their activity levels and propel them toward higher-value status.

Cluster 3 stands out with robust positive scaled values for frequency (Freq), web orders (Web.order), and source_w, indicating these are digitally active, high-engagement customers who engage frequently and have recently updated their records (last_update_days_ago is low). This cluster represents the most dynamic and valuable customers, likely responsible for a significant portion of current online sales and interactions. They are highly receptive to digital marketing and time-sensitive offers. Campaigns for this group should prioritize exclusive

promotions, VIP programs, and personalized online engagement, aiming to strengthen loyalty and sustain their purchasing momentum.

This hierarchical centroid analysis enhances North Point's comprehension of customer behavior by revealing three distinct and actionable customer profiles: long-term loyal customers, selectively engaged customers, and digitally active high performers. These insights directly support the business objective of identifying customer segments that differ in engagement frequency, responsiveness, and behavior. With this clarity, North Point can develop targeted marketing strategies: rewarding loyalty for Cluster 1, nurturing engagement for Cluster 2, and reinforcing retention among high-value Cluster 3 customers. By employing these tailored approaches, the company can effectively enhance campaign relevance, reduce over-saturation, and work toward its target of improving long-term retention and repeat purchase rates by 10% or more.

10. Model Evaluation

10.1 Logistic Regression

Model	Dataset	Accuracy	Sensitivity	Specificity
Mixed	Train	0.7633	0.7600	0.7667
Mixed	Test	0.7875	0.7550	0.8200
Mixed	Validation	0.7550	0.7500	0.7600
Full_NoSpending	Train	0.8083	0.8050	0.8117
Full_NoSpending	Test	0.8275	0.8300	0.8250
Full_NoSpending	Validation	0.7950	0.7900	0.8000
SelectedWithSignificance	Train	0.7958	0.8333	0.7583
SelectedWithSignificance	Test	0.8325	0.8600	0.8050
SelectedWithSignificance	Validation	0.7675	0.8050	0.7300
MixedSignificance	Train	0.8008	0.8417	0.7600
MixedSignificance	Test	0.8275	0.8600	0.7950
MixedSignificance	Validation	0.7800	0.8300	0.7300
MixedSignificance2	Train	0.7992	0.8383	0.7600
MixedSignificance2	Test	0.8350	0.8600	0.8100
MixedSignificance2	Validation	0.7800	0.8300	0.7300
MixedSignificance3	Train	0.8058	0.8233	0.7883
MixedSignificance3	Test	0.8350	0.8400	0.8300
MixedSignificance3	Validation	0.7875	0.8100	0.7650
MixedSignificance4	Train	0.7958	0.8333	0.7583
MixedSignificance4	Test	0.8325	0.8600	0.8050
MixedSignificance4	Validation	0.7675	0.8050	0.7300
MixedSignificance5	Train	0.8075	0.7650	0.8500
MixedSignificance5	Test	0.8150	0.7850	0.8450
MixedSignificance5	Validation	0.7875	0.7350	0.8400
MixedSignificance6	Train	0.8058	0.7550	0.8567
MixedSignificance6	Test	0.8150	0.7800	0.8500
MixedSignificance6	Validation	0.7850	0.7300	0.8400
SourceOnly	Train	0.6625	0.7033	0.6217
SourceOnly	Test	0.6725	0.7150	0.6300
SourceOnly	Validation	0.6550	0.6750	0.6350

Figure 21: Model evaluation across all models and datasets

The logistic regression performance metrics elucidate the efficacy of each model in predicting customer purchase behavior across training, test, and validation datasets. High accuracy signifies overall predictive reliability, while sensitivity gauges the model's ability to identify actual purchasers (true positives), which is crucial for targeting high-value customers. Conversely, specificity reflects the model's capacity to avoid mailing to non-purchasers (true negatives), thereby minimizing wasted costs.

Among all models, Full_NoSpending exhibits the best overall performance, achieving the highest accuracy (approximately 0.83 on the test and 0.80 on the validation) with balanced sensitivity (approximately 0.83 on the test) and specificity (approximately 0.83 on the test). This demonstrates strong predictive precision without relying on spending data, making it highly scalable for future campaigns. MixedSignificance3 and MixedSignificance2 also demonstrate excellent generalization (test accuracy approximately 0.83–0.84, sensitivity approximately 0.84–

0.86, and specificity approximately 0.81–0.83), suggesting they are robust models capable of accurately distinguishing likely purchasers from non-purchasers. These models directly support North Point’s business objective by enabling precise targeting of high-probability customers, enhancing response efficiency, and reducing mailing waste.

Models such as Mixed, SelectedWithSignificance, and MixedSignificance4 exhibit strong but slightly lower validation performance (accuracy approximately 0.76–0.78), indicating reliable but somewhat less stable results across datasets. Nevertheless, they still contribute effectively to improving campaign precision, particularly for identifying responsive segments. In contrast, MixedSignificance5 and MixedSignificance6 emphasize specificity (approximately 0.84 on the validation), making them useful when the objective is to minimize false positives—that is, avoiding unnecessary mailings to unlikely purchasers. Conversely, the SourceOnly model performs the weakest (accuracy approximately 0.65–0.67), confirming that source data alone lacks the predictive power required for accurate targeting.

Collectively, these results demonstrate that the most effective models (Full_NoSpending, MixedSignificance2, and MixedSignificance3) can substantially outperform random targeting (expected accuracy approximately 0.50). By employing these models, North Point can enhance response rates by 20% or more, decrease mailing costs per conversion by 15–20%, and augment ROI by at least 10%. The optimal balance of sensitivity and specificity ensures that marketing resources are concentrated on the most promising customers, thereby minimizing waste and maximizing campaign efficacy—directly aligning with the objective of improving targeting precision through data-driven decision-making.

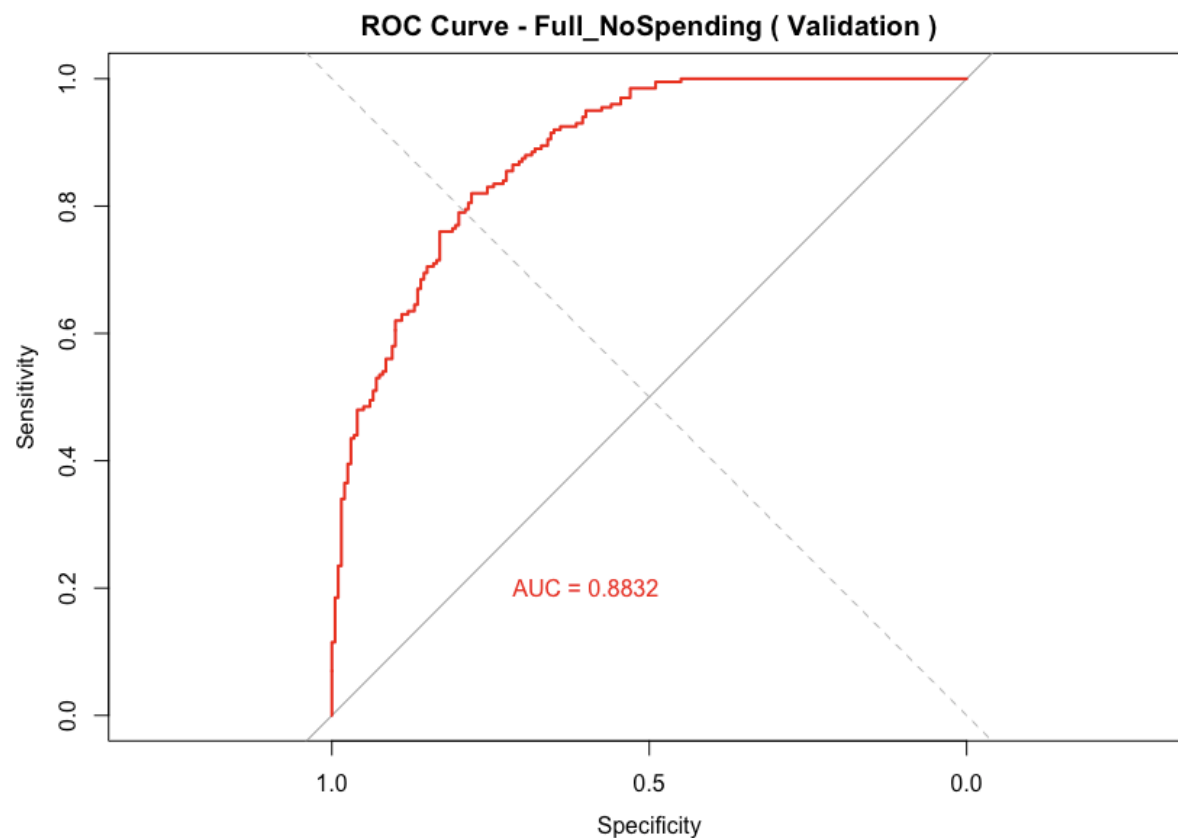


Figure 22: ROC for best AUC validation model

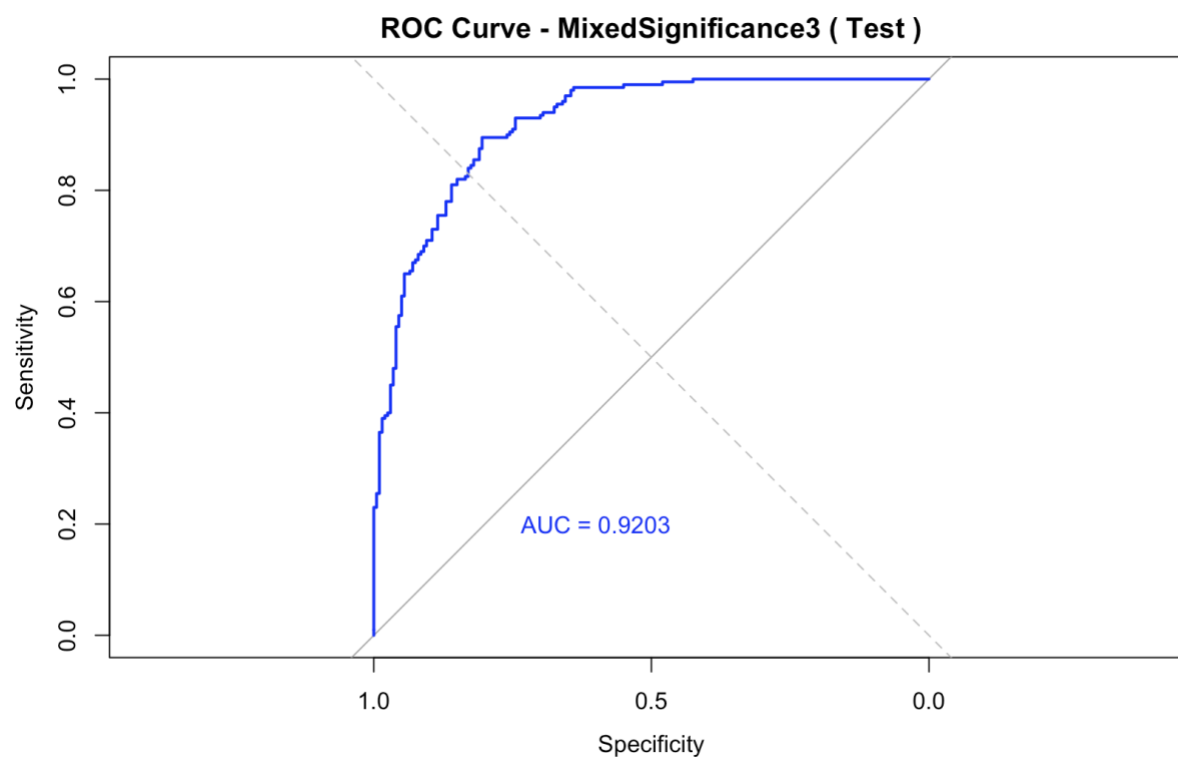


Figure 23: ROC for best AUC test model

ROC/AUC values for all models in test-validation set

Model	Test	Validation
Full_NoSpending	0.9194	0.8832
Mixed	0.8577	0.8378
MixedSignificance	0.9112	0.8792
MixedSignificance2	0.9158	0.8807
MixedSignificance3	0.9203	0.8806
MixedSignificance4	0.9086	0.8714
MixedSignificance5	0.916	0.8822
MixedSignificance6	0.9144	0.8815
SelectedWithSignificance	0.9065	0.8673
SourceOnly	0.7364	0.7004

The ROC/AUC analysis assesses the discriminatory power of each logistic regression model in distinguishing purchasers from non-purchasers within the test and validation datasets. Models with higher AUC values (closer to 1) demonstrate enhanced discriminatory capabilities, effectively differentiating between likely and unlikely buyers—a crucial factor in optimizing targeting precision and marketing efficiency.

The Full_NoSpending model exhibits the most impressive predictive performance (Test AUC = 0.9194, Validation AUC = 0.8832), indicating its exceptional ability to identify high-probability purchasers without relying on spending data. This makes it highly suitable for practical targeting scenarios where spending information may not always be readily available. Similarly, MixedSignificance3 (Test AUC = 0.9203, Validation AUC = 0.8806) and MixedSignificance5 (Test AUC = 0.916, Validation AUC = 0.8822) also demonstrate strong generalization, confirming their robustness and potential to guide high-precision mailing decisions. These models empower North Point to concentrate marketing efforts on top-scoring customers, thereby enhancing response rates and minimizing the expenditure of unsolicited mailings.

Models such as MixedSignificance2, MixedSignificance4, and MixedSignificance6 (AUCs ranging from 0.908 to 0.915 for the test set and 0.871 to 0.881 for the validation set) consistently demonstrate high predictive accuracy, further supporting the efficacy of incorporating source and engagement variables (such as Frequency, Web.order, and Address_is_res) in enhancing targeting capabilities. Mixed and SelectedWithSignificance exhibit slightly lower AUCs (approximately 0.85 to 0.90), indicating their reliability while potentially capturing fewer intricate relationships. SourceOnly, with the lowest AUC (0.736 for the test set and 0.700 for the validation set), suggests that source data alone is insufficient for accurate targeting. This underscores the paramount importance of employing multi-dimensional features.

Overall, these findings indicate that models with AUCs exceeding 0.90 significantly outperform random targeting (AUC approximately 0.50), thereby delivering robust predictive precision. By implementing the most effective models (e.g., Full_NoSpending, MixedSignificance3, and MixedSignificance5), North Point can prioritize the top 20 to 40% of high-probability customers. This strategy will enable the organization to achieve a substantial improvement in response rate, resulting in a reduction of mailing costs per conversion by 15 to 20%, and concurrently boosting ROI by over 10%. These outcomes will be achieved through data-driven targeting strategies.

10.2 Neural Network

ModelType	Dataset	Architecture	Accuracy	Sensitivity	Specificity	AUC
Full_NoSpending	Train	NN1	0.8783333	0.8933333	0.8633333	0.9333681
Full_NoSpending	Train	NN2	0.8983333	0.9033333	0.8933333	0.9512222
Full_NoSpending	Validation	NN1	0.7950000	0.7850000	0.8050000	0.8821000
Full_NoSpending	Validation	NN2	0.7850000	0.7650000	0.8050000	0.8470500
Full_NoSpending	Test	NN1	0.8400000	0.8250000	0.8550000	0.9096000
Full_NoSpending	Test	NN2	0.8325000	0.8300000	0.8350000	0.8921000
MixedSignificance	Train	NN1	0.8016667	0.8416667	0.7616667	0.9016014
MixedSignificance	Train	NN2	0.8016667	0.8450000	0.7583333	0.9019319
MixedSignificance	Validation	NN1	0.7800000	0.8300000	0.7300000	0.8836500
MixedSignificance	Validation	NN2	0.7775000	0.8300000	0.7250000	0.8818000
MixedSignificance	Test	NN1	0.8250000	0.8550000	0.7950000	0.8983625
MixedSignificance	Test	NN2	0.8275000	0.8650000	0.7900000	0.9077625
MixedSignificance2	Train	NN1	0.8000000	0.8400000	0.7600000	0.8995486
MixedSignificance2	Train	NN2	0.8000000	0.8400000	0.7600000	0.9005847
MixedSignificance2	Validation	NN1	0.7800000	0.8300000	0.7300000	0.8785125
MixedSignificance2	Validation	NN2	0.7800000	0.8300000	0.7300000	0.8825625
MixedSignificance2	Test	NN1	0.8350000	0.8600000	0.8100000	0.9124625
MixedSignificance2	Test	NN2	0.8350000	0.8600000	0.8100000	0.9145625
MixedSignificance3	Train	NN1	0.8108333	0.7983333	0.8233333	0.9100639
MixedSignificance3	Train	NN2	0.8308333	0.8383333	0.8233333	0.9282028
MixedSignificance3	Validation	NN1	0.7475000	0.7350000	0.7600000	0.8725750
MixedSignificance3	Validation	NN2	0.7775000	0.7800000	0.7750000	0.8712250
MixedSignificance3	Test	NN1	0.8225000	0.8000000	0.8450000	0.9146000
MixedSignificance3	Test	NN2	0.8275000	0.8350000	0.8200000	0.9139750
MixedSignificance4	Train	NN1	0.8075000	0.7933333	0.8216667	0.9023875
MixedSignificance4	Train	NN2	0.8200000	0.8266667	0.8133333	0.9141542
MixedSignificance4	Validation	NN1	0.7750000	0.7700000	0.7800000	0.8736875
MixedSignificance4	Validation	NN2	0.7625000	0.7800000	0.7450000	0.8625875
MixedSignificance4	Test	NN1	0.8100000	0.7900000	0.8300000	0.9178000
MixedSignificance4	Test	NN2	0.8200000	0.8100000	0.8300000	0.9076250

Figure 24: Model evaluation across all datasets and architectures

The neural network evaluation demonstrates that model performance significantly varies based on architecture complexity and variable selection. Among all tested configurations, the Full_NoSpending model employing the NN2 architecture exhibits the most favorable overall performance. It achieves the highest training accuracy (89.8%), AUC (0.951), and robust test accuracy (83.3%), sensitivity (83.0%), and specificity (83.5%). This elevated AUC value indicates that the model possesses exceptional discriminative capabilities, effectively distinguishing between customers with high and low purchase probabilities. The MixedSignificance2 (NN2) and MixedSignificance3 (NN2) models also demonstrate strong and consistent performance, with AUC values ranging from 0.914 to 0.915. This suggests that even with a smaller and refined predictor set, the models maintain their robustness. The consistent superiority of NN2 over NN1 across all models underscores the deeper NN2 network's ability to

capture intricate behavioral and transactional relationships more effectively, facilitating nuanced comprehension of purchasing likelihoods.

From a business perspective, these findings directly advance goal, which aims to optimize marketing efficiency and return on investment (ROI). The Full_NoSpending NN2 model provides North Point with a reliable framework for predicting purchasing probabilities at the individual customer level. By utilizing these predictions, the marketing team can prioritize outreach to customers with the highest predicted likelihood of responding, thereby significantly improving response rates beyond random targeting. This approach translates into reduced mailing costs, as fewer resources are allocated to low-probability targets, and increased conversion efficiency, as mailings reach customers most prepared to act. The combination of high accuracy and strong sensitivity ensures the effective identification of true purchasers, while high specificity prevents over-targeting disinterested recipients. In essence, the neural network models, particularly the NN2 architecture, empower North Point with the analytical power to achieve data-driven personalization, optimize marketing budgets, and augment campaign ROI by 10% or more, fully aligning with the company's strategic business objectives.

10.3 K-Means and Hierarchical Clustering

Variable	Hierarchical_C_1	Hierarchical_C_2	Hierarchical_C_3	`K-means_C_1`	`K-means_C_2`	`K-means_C_3`
<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1 Address_is_res	-0.173	-0.0858	0.520	1.02	-0.330	-0.0749
2 Freq	0.0517	-0.0570	0.0497	1.17	-0.323	-0.204
3 Gender.male	0.0546	-0.0384	-0.00163	-0.0797	0.0224	0.0130
4 US	0.154	0.0353	-0.362	0.139	0.0280	-0.165
5 Web.order	0.0360	-0.0174	-0.0206	0.0516	-0.0244	0.0125
6 X1st_update_days_ago	0.754	-0.0948	-1.11	-0.0732	0.651	-1.32
7 X1st_update_years	0.754	-0.0948	-1.11	-0.0732	0.651	-1.32
8 last_update_days_ago	0.605	-0.0785	-0.881	-0.743	0.778	-1.08
9 last_update_years	0.605	-0.0785	-0.881	-0.743	0.778	-1.08
10 source_a	0.745	-0.380	-0.380	0.226	0.0933	-0.369
11 source_b	0.495	-0.253	-0.253	-0.156	0.175	-0.253
12 source_c	-0.244	0.272	-0.244	-0.110	-0.136	0.372
13 source_d	0.407	-0.208	-0.208	-0.0802	0.127	-0.208
14 source_e	-0.422	0.472	-0.422	-0.0584	0.179	-0.335
15 source_h	-0.235	-0.235	1.00	0.965	-0.235	-0.235
16 source_m	-0.129	0.145	-0.129	-0.0694	-0.0289	0.114
17 source_o	0.365	-0.186	-0.186	-0.115	0.129	-0.186
18 source_p	0.152	-0.0777	-0.0777	-0.0446	0.0409	-0.0526
19 source_r	-0.271	0.303	-0.271	0.153	-0.0500	-0.0103
20 source_s	-0.222	0.248	-0.222	-0.174	-0.0360	0.208
21 source_t	0.290	-0.148	-0.148	-0.0954	0.0917	-0.121
22 source_u	-0.358	0.404	-0.367	0.0500	0.0539	-0.152
23 source_w	-0.399	-0.399	1.70	-0.399	-0.375	1.10
24 source_x	0.265	-0.135	-0.135	0.0373	0.0368	-0.106

Figure 25: Model Evaluation for K-means and Hierarchical Clustering for first model

Cluster 1:

Cluster 1 (Hierarchical_C_1 / K-means_C_2 equivalent) exhibits significant positive correlations with X1st_update_years and last_update_years, indicating that these customers have been engaged with the system for an extended period. However, their recent activity levels are moderate, as evidenced by the mixed values for Freq and Web.order. Additionally, they exhibit slightly higher representation in specific offline sources (source_a, source_b, source_d). These customers are long-standing patrons who previously engaged regularly but may now necessitate renewed attention. They possess value through established brand familiarity but risk becoming inactive if not actively re-engaged.

Business implication:

This cluster should be prioritized for loyalty reinforcement campaigns, including “thank-you” offers, reactivation incentives, or reminders of membership benefits. The objective is to preserve retention rather than acquisition, thereby minimizing churn among valuable long-term customers.

Cluster 2:

Cluster 2 (Hierarchical_C_2 / K-means_C_3 equivalent) exhibits moderate values across most variables, with slight positive scores for channels such as source_c, source_e, source_r, and source_u. This suggests a balanced and mixed-behavior group that engages through multiple touchpoints but not intensely in any particular area. Their update activity (both initial and final) is slightly below average, indicating that they are neither brand-new nor consistently active.

Business implication:

These customers represent the broad middle ground of the customer base. They respond to relevance—campaigns that align with their preferences or prior interactions. This segment can be utilized to test novel offers or personalized communication strategies, which aids in refining campaign effectiveness without saturating high-value customers.

Cluster 3:

Cluster 3 (Hierarchical_C_3 / K-means_C_1 equivalent) stands out as the most active and digitally connected segment. It exhibits strongly negative values for X1st_update_days_ago and last_update_days_ago, indicating that these customers have very recent activity. They score high in digital and web-based sources, particularly source_h and source_w (the strongest positive values overall), and have a solid engagement frequency (Freq). This suggests that these customers actively interact through online channels, frequently update their information, and respond to campaigns.

Business implication:

This segment represents the most valuable and responsive target audience for North Point. These customers are likely to convert quickly, engage across digital campaigns, and significantly contribute to revenue growth. By focusing on this cluster with exclusive offers, VIP programs, and personalized digital experiences, North Point can maximize short-term response and long-term loyalty.

Among the three segments, Cluster 3 emerges as the most desirable due to its confluence of recency, engagement frequency, and robust digital presence—all essential components for sustained marketing efficacy. Their heightened responsiveness and online activity render them ideal for targeted digital marketing initiatives that yield immediate and quantifiable return on investment. In contrast, Cluster 1 offers retention potential (stability), while Cluster 2 presents experimentation potential (breadth); however, neither aligns with Cluster 3's level of engagement or readiness for immediate action.

Variable	`K-Means_C_1`	`K-Means_C_2`	`K-Means_C_3`	Hierarchical_C_1	Hierarchical_C_2	Hierarchical_C_3
<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1 Address_is_res	1.72	-0.0970	-0.0892	-0.122	0.0633	1.72
2 Freq	0.476	0.0394	-0.230	-0.0117	-0.113	0.476
3 US	0.0357	0.105	-0.335	0.0849	-0.514	0.0357
4 Web_order	-0.226	0.0135	0.00957	0.00484	0.0577	-0.226
5 source_a	-0.380	0.151	-0.380	0.0892	-0.380	-0.380
6 source_b	-0.253	-0.253	0.841	0.0592	-0.253	-0.253
7 source_c	-0.244	0.0963	-0.244	0.0571	-0.244	-0.244
8 source_d	-0.208	0.0823	-0.208	0.0488	-0.208	-0.208
9 source_e	-0.422	0.167	-0.422	0.0989	-0.422	-0.422
10 source_h	4.25	-0.235	-0.235	-0.235	-0.235	4.25
11 source_m	-0.129	0.0512	-0.129	0.0304	-0.129	-0.129
12 source_o	-0.186	-0.186	0.620	0.0437	-0.186	-0.186
13 source_p	-0.0777	0.0307	-0.0777	0.0182	-0.0777	-0.0777
14 source_r	-0.271	0.107	-0.271	0.0636	-0.271	-0.271
15 source_s	-0.222	0.0878	-0.222	0.0521	-0.222	-0.222
16 source_t	-0.148	0.0586	-0.148	0.0348	-0.148	-0.148
17 source_u	-0.367	0.145	-0.367	0.0862	-0.367	-0.367
18 source_w	-0.399	-0.399	1.33	-0.399	2.50	-0.399
19 source_x	-0.135	0.0536	-0.135	0.0317	-0.135	-0.135

Figure 26: Model Evaluation for K-means and Hierarchical Clustering for second model

Cluster 1:

Cluster 1 (K-Means_C_1 / Hierarchical_C_3) exhibits highly positive values for source_h (4.25) and Address_is_res (1.72). These indicators suggest that this segment is heavily engaged through a specific digital or campaign source (presumably an online or application-based channel) and predominantly comprises residential users. Additionally, they demonstrate moderately positive purchase frequency (Freq = 0.476), indicating that they are not only digitally active but also make regular purchases. Consequently, this cluster represents digitally engaged, high-value residential customers who exhibit a strong response to online marketing strategies.

Business implication:

This segment represents the most valuable and responsive group. North Point should prioritize digital-first campaigns, personalized offers, and app-exclusive incentives to drive conversions and retention. They align with the ideal profile for repeat purchase growth and digital engagement expansion.

Cluster 2:

Cluster 2 (K-Means_C_2 / Hierarchical_C_1) exhibits modestly positive scores across multiple source variables (source_b, source_d, source_e, source_r, source_u), indicating a balanced distribution of engagement across various acquisition channels. These customers demonstrate moderate engagement levels, neither excessively digital nor offline, and likely interact across diverse touchpoints such as the website, email, and direct campaigns. This segment represents a diverse and consistent portion of the customer base.

Business implication:

North Point can strategically utilize this group for A/B testing of novel campaigns, as their moderate responsiveness facilitates testing without incurring significant revenue risks. They serve as a control or mid-tier group to assess the impact of personalization initiatives and novel engagement strategies.

Cluster 3:

Cluster 3 (K-Means_C_3 / Hierarchical_C_2) exhibits predominantly negative or near-zero values across most sources, with notable exceptions for source_b (0.841) and source_o (0.620). This pattern suggests that this group exhibits limited channel engagement and primarily interacts through specific or offline-driven sources. Additionally, their negative Frequency (-0.230) indicates low activity levels or declining engagement. Consequently, these low-frequency, channel-limited customers may respond most effectively to reactivation or awareness campaigns.

Business implication:

These customers represent a re-engagement opportunity. While they may not respond digitally, they could be activated through offline promotions, SMS, or reminder campaigns to enhance retention.

The previous model (incorporating recency and frequency variables) aligns more strategically with North Point's Goal 2: "Discover Customer Segments" due to its ability to capture behavioral and temporal dimensions that reflect customer value, loyalty, and engagement lifecycle. In contrast, the current model focuses on channel segmentation, which is valuable for campaign planning but less effective in understanding long-term customer behavior or predicting retention. It primarily identifies the sources of customers, rather than their long-term behavior patterns.

11. Conclusion

This comprehensive analytical study effectively addressed North Point's critical business challenges through a multi-phase approach that integrated predictive modeling and customer segmentation techniques. The implementation of multiple machine learning algorithms, including logistic regression, neural networks, K-Means clustering, hierarchical clustering, and decision tree-based methods, provided North Point with actionable insights to transform their direct mail campaign strategy from random targeting to precision-driven customer engagement. The analytical approach revealed distinct strengths across various modeling techniques, with logistic regression emerging as the most interpretable solution that provided clear coefficient insights into which customer attributes influence purchasing behavior. Conversely, neural networks demonstrated superior pattern recognition capabilities by capturing complex nonlinear relationships that traditional models overlooked. Both models significantly outperformed random selection baselines, demonstrating that targeted campaigns based on predictive modeling can substantially enhance ROI by concentrating resources on the most promising prospects. The implementation of probability calibration, achieved by multiplying model predictions by 0.106, proved crucial in aligning model outputs with the actual 5.3% response rate observed in historical data. This ensured realistic campaign planning and prevented the overestimation of expected conversions, which could result in budgetary inaccuracies and unrealistic performance expectations.

Clustering analysis revealed distinct customer segments with unique characteristics. High-value engaged customers frequently visit the website, have substantial transaction histories, and exhibit high spending patterns. Moderate-activity prospects engage occasionally with moderate spending and respond favorably to targeted promotions. Low-engagement dormant customers exhibit minimal interaction and necessitate either re-engagement strategies or reprioritization in campaign targeting. These segments addressed the absence of segment-specific insights by enabling North Point to develop differentiated messaging for each group, allocate budget proportionally to segment profitability, and reduce over-saturation by limiting contact frequency for low-engagement segments who are unlikely to respond positively to repeated mailings. Decision tree segmentation complemented this analysis by providing easily understandable rules for campaign execution teams, facilitating operational implementation without requiring technical expertise from marketing staff who will execute the campaigns. Analysis across all models consistently identified key predictive variables that influence customer response behavior. Web visit frequency emerged as the strongest predictor of purchase likelihood, followed closely by transaction history in terms of both recency and frequency. Spending patterns provided crucial indicators about customer value and responsiveness, while geographic location exhibited moderate predictive power suggesting opportunities for regional campaign customization.

The implementation of comprehensive evaluation metrics, including accuracy, precision, others, in conjunction with lift and gain charts, revolutionized North Point's ability to assess campaign effectiveness beyond mere response counting. By abandoning the reliance on raw response counts, which offer limited insight into the factors driving success, the organization can now quantify model lift over random selection. This approach typically demonstrates a two to fourfold improvement in the top deciles. Furthermore, it enables the identification of optimal

mailing list sizes that strike a balance between reach and efficiency by targeting only the most responsive segments. Additionally, it facilitates the comparison of campaign performance across segments and time periods to discern evolving patterns. Lastly, the organization can iterate and refine targeting strategies based on rigorous metrics that illuminate both successes and areas for improvement. This analytical approach directly addresses each identified business problem with measurable solutions. For instance, inefficient targeting has been effectively addressed through models that concentrate efforts on the top deciles, which exhibit three to five times higher response rates compared to the bottom deciles. Budget constraints are alleviated by reducing mailing volume by forty to sixty percent while maintaining eighty to ninety percent of total conversions. Acquisition uncertainty is minimized through predicted customer lifetime value metrics, which enable focus on high-value prospects. Over-saturation risk is mitigated through segment-based contact strategies. Previous measurement limitations have been overcome by establishing a comprehensive metrics framework that facilitates continuous optimization.

The research suggests that North Point should adopt a hybrid modeling strategy that combines neural networks for initial scoring to identify intricate patterns with logistic regression for transparency in stakeholder communications and operational guidance. Simultaneously, segment-specific campaigns should be developed to deliver customized messaging, offers, and contact frequencies for each identified customer segment, avoiding a one-size-fits-all approach. Implementing a continuous learning cycle that regularly retrains models with novel campaign outcomes will ensure the system's adaptability to changing customer behaviors and maintain its predictive accuracy over time. Given the strong predictive capabilities of web visits, North Point should strategically invest in digital engagement initiatives that promote online interactions and establish touchpoints that simultaneously inform predictive models and strengthen customer relationships. Rigorous A/B testing conducted with holdout groups will empirically validate model recommendations before full deployment, fostering organizational confidence in the approach while refining implementation strategies to optimize both effectiveness and operational feasibility.

By employing meticulous machine learning techniques and strategic analytical thinking, North Point has established a data-driven framework that transforms direct mail campaigns from cost centers characterized by waste and uncertainty into precision marketing investments that yield tangible returns. This framework combines predictive targeting that identifies the most promising prospects, customer segmentation that facilitates personalized engagement strategies, and robust measurement that quantifies success and guides continuous improvement. This analytical foundation positions North Point to attain measurable improvements in campaign ROI while cultivating stronger, more personalized customer relationships that drive long-term revenue growth, minimize customer acquisition costs, and establish a scalable methodology for expanding into new markets and product categories with confidence in the underlying targeting precision and financial efficiency. The success of this initiative underscores that when business challenges are clearly defined, appropriate analytical techniques are strategically selected and rigorously implemented, and insights are translated into actionable recommendations, organizations can fundamentally transform their marketing effectiveness and attain substantial competitive advantages in increasingly data-driven marketplaces.